CHAPTER

7

Diagnosis and Diagnostic Aids

The main goal of restorative dentistry (functional occlusion and pleasing esthetics) can only be achieved with sound foundation, which in any treatment regime is 'accurate diagnosis'. Dental treatment in any form should never be embarked upon until the diagnosis is established. Diagnosis is the most critical aspect of patient care since adequate treatment planning depends on accurate diagnosis. The process of diagnosis involves collection of information, observing signs and symptoms, thorough clinical examination and accurate interpretation of objective testing. Various scientific devices can be used to gather information and the thoughtful interpretation of the collected information help arriving at diagnosis. An accurate diagnosis is a result of the synthesis of scientific knowledge, clinical experience, intuition and also common sense.

Definition

Diagnosis literally means the determination and judgement of variations from the 'normal'. It is defined as *"the utilization of scientific knowledge for identifying a disease process and to differentiate it from other disease processes"*.

The word 'diagnosis' is a Greek word, meaning specifically discriminating/distinguishing between two possibilities. In the present concept, this is interpreted as 'differential diagnosis'.

Oxford dictionary defines diagnosis as *"the identification of nature of an illness/problem by examination of the symptoms"*.

The accepted definitions of diagnosis, as mentioned by other authors, are:

McGhee: "The correct determination, discriminative estimation and logical appraisal of conditions found during examination as evidenced by distinctive marks, signs and symptoms that are characteristic of health or disease".

Grossman: "The process of identifying a disease by its signs, symptoms and results of various diagnostic procedures".

Diagnostic tests provide varying degree of certainty. Usually, the test is interpreted as positive or negative; hence, the disease is either present or absent. However, in certain cases, the diagnostic tools are misleading, i.e. the test can be positive in the absence of disease (false positive) or negative in the presence of disease (false negative).

Diagnostic tests are usually assessed in terms of specificity and sensitivity. *Specificity* refers to the ability of the test or observation to clearly differentiate one disease from the other. Specificity is the ability to rule out the lesion when it is truly absent. *Sensitivity* refers to the ability of a test or observation to detect the disease, whenever it is truly present. The sensitivity and the specificity of a test should be considered when the decision has to be made as to which test should be preferred for a particular disease.

It is well known that the process of diagnosis starts the moment patient enters the clinic. Patient's overall assessment, way of dressing, neatness of clothes, gait and general well-being should be noticed at the first instance. Excitement and apprehension on the face of the patient should also be observed. The patient is to be psychologically assessed.

The operator should have sympathetic ears to hear, eyes to see, hands to feel, nose to smell, brain to analyze and clinical experience to derive at proper diagnosis. He/she should have:

- Knowledge of disease process
- Interest in the subject
- Concern for patient care
- Curiosity to find the details
- Intuition to suspect unusual
- Patience to finalize the diagnosis

CARIES DIAGNOSIS

Diagnosis in restorative phase is mainly the detection of caries. Other features, such as discoloration, fracture, tooth substance loss, etc. can be diagnosed by visual examination only. In a few cases, pulp

vitality testing may have to be undertaken to ascertain status of pulp in a particular tooth.

Requisites of Diagnostic Tests

The caries diagnostic test must be:

- Accurate
- Sensitive and specific
- Reproducible and reliable
- Should not transfer any micro-organism from affected area to the unaffected area.
- Time and cost should be manageable.

DIAGNOSTIC TOOLS

Various methods (tools) to diagnose caries are:

1. Visual examination
2. Tactile examination
3. Radiographic methods
   a. Conventional radiography
      i. Intraoral periapical radiography
      ii. Bitewing radiography
   b. Xeroradiography
   c. Modified radiographic techniques
      i. Direct digital radiography
      ii. Computer image analysis
      iii. Subtraction radiography
      iv. Computed tomography (CT) scan
      v. Microcomputed tomography (micro-CT) scan
      vi. Cone beam computed tomography (CBCT)
      vii. Magnetic resonance imaging (MRI)
4. Electric resistance (electrical conductance and impedance)
   a. Electronic caries meter (ECM)
   b. Caries meter-L
   c. Vanguard electronic caries detector
   d. Electronic caries meter (ECM)
   e. Electrical impedance tomography
   f. Alternating current impedance spectroscopy (CarieScan)
5. Optical detection methods
   a. Optical caries monitor
   b. Fiberoptic transillumination (FOTI)
   c. Digital imaging fiberoptic transillumination (DIFOTI)
   d. Ultraviolet illumination
   e. Transillumination with near infrared light
   f. Laser induced fluorescence

   Devices (with camera) using fluorescence principles
      i. Quantitative light-induced fluorescence (QLF)
      ii. Light-induced fluorescence evaluation—diagnosis and treatment (LIFEDT) concept
      iii. Newer cameras

   Devices (without camera) using fluorescence principles
      i. DIAGNOdent
   g. Visible lumniscent spectroscopy
   h. Terahertz imaging (TI)
   i. Multiphoton imaging (MPI)
   j. Optical Coherence Tomography (OCT)
   k. Polarized Raman spectroscopy (PRS)
   l. Midwest caries ID (LED technology)
   m. Frequency-Domain infrared photo thermal radiometry and modulated luminescence (PTR/LUM)
   n. Time-correlated single-photon counting fluorescence lifetime imaging
6. Endoscopy/videoscopy
7. Dye penetration methods
8. Miscellaneous
   a. Laser
   b. Infrared thermography
   c. Ultrasonic imaging
   d. Chemical analysis
   e. Microradiography
      - Transverse microradiography
      - Longitudinal microradiography
      - Wavelength independent microradiography
   f. Energy dispersive X-ray element analysis
   g. Microhardness analysis
   h. Iodine absorptiometry
   i. Polarized light microscopy

1. Visual Examination

Visual examination needs sufficient time and patience. The examination involves visually looking for surface roughness, opacification, discoloration and cavitation. The teeth are cleaned and dried with compressed air and illuminated under adequate light source. The limitations of using this method is that discoloration of the pits and fissures, a common finding in normal healthy adult teeth (extrinsic stains), may simulate discoloration of caries. The individual's past history and the habits can help differentiating between these two discolorations.

2. Tactile Examination

Tactile examination includes determining roughness/softness of the tooth surface with a sharp explorer. Both penetration and resistance to removal of an explorer tip (catch) have been interpreted as an evidence of caries. Probing is an age-old diagnostic aid; however, it has been criticized and questioned because of several reasons viz:

a. May transmit cariogenic bacteria from one site to another (one probe is used to examine different sites).

b. May produce irreversible traumatic defects in enamel, which further become susceptible to caries attack.
c. May not add any information to the visual examination.
d. The explorer tip may bind in a fissure because of reasons other than caries. The reasons can be:
   i. Shape of the fissure: Small/rounded pits/fissures can bind the tip.
   ii. Sharpness of an explorer: A sharp explorer has a diameter of 200 microns at its tip and pressing the explorer may push the tip into enamel.
   iii. Force of application: Heavy pressure may lead to binding of the probe tip into enamel fissures.

Dental floss and tooth separators are also employed to detect proximal caries. After the tooth has been thoroughly cleaned, floss is inserted through the contact area and dragged occlusally against one proximal surface. If it shreds, one can suspect a proximal cavity; however, overhanging restorations on the proximal side may present similar features. This is to be looked into prior to flossing. Tooth separators and wedges aid in detecting the proximal caries visually by separating the teeth.

The commonly utilized visual-tactile criteria for detection of caries as proposed by various authors in the past are tabulated in Table 7.1.

Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective

Jackson D (1950)	There should be positive and clear evidence of enamel dissolution before the same is diagnosed as caries. 1. A clear mirror and sharp probe must be used for tactile examination. (Ash's Sickle Probe No. 54 is used for pit and fissure lesions; whereas, Ash's Probe No. 12 is used for proximal lesions.) 2. The tooth must be dried. 3. A pit/fissure is counted as carious, if, with a little pressure, the point sticks and requires a definite pull to be removed (doubtful ones are not included). 4. Stained pits/fissures are not counted as carious; unless they satisfy this test. 5. Stained or opaque light areas on other smooth surfaces are not counted as carious, unless they show evidence of enamel dissolution. 6. Proximal lesions are considered 'carious', if Ash's No. 12 probe catches a roughened surface or a discontinuity. 7. Arrested caries is counted as carious, and exposed dentin in hypoplastic teeth is also counted as carious (only if there is positive evidence of softening).
Parfitt GJ (1954)	A slight discoloration with loss of luster on the enamel surface is considered as the first sign of caries. Grade 1: Slight discoloration with loss of luster of the enamel surface Grade 2: Surface is roughened and pitted; a condition, which can be detected by explorer point. Grade 3: Further penetration and loss of tissue, causing pitting to reach the dentin Grade 4: Loss of dentin and cavitation
Backers-Derks, et al (1961)	The clinical examination by mirror and explorer to detect proximal caries was discontinued because of poor accuracy. The teeth were cleaned and dried with compressed air and examined under transmitted light. Caries was estimated in 4 different grades. Grade I signifies a minute black line at the bottom of the fissure. Grade II, also a white zone along the margins of the fissure. Grade III denotes the smallest perceptible break in the continuity of the enamel (cavity) with or without undermined margins. Grade IV is a large cavity more than 3.0 mm wide.
McHugh, et al. (1964)	Teeth were counted as carious, if a probe stuck definitely in a pit/fissure after applying gentle pressure; designated as 'sticky fissure'. In addition, each carious cavity was given a 'penetration score' on the following basis: Score 1 = Sticky fissure Score 2 = Fissure or smooth surface cavity with softness at base and staining or opacity of the enamel Score 3 = Cavity with obvious dentin involvement in the proximal surfaces Score 4 = Cavity with pulp involvement
Marthaler TM (1966)	Emphasis was given to visual examination. (Probe was used only in case of doubt). Grade 1: Slightly brown narrow line (on smooth surfaces Class V) or white spot with hard surface, smallest extent not exceeding 2.0 mm Grade 2: Clearly brown or black line (on Class V lesions) or white spot, smallest extent exceeding 2.0 mm; For Class III lesions (proximal of anterior teeth), the lesion has a dark brown discolored surface.

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Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective *(Contd...)*

	Grade 3: Cavity, discontinuity of the enamel surface Grade 4: Cavity with the narrowest extent of the entrance broader than 2.0 mm
Moller IJ (1966)	Caries on smooth surfaces (buccal and lingual) Grade I: A white opaque spot that keeps its luster after a short (3 seconds) period of drying Grade II: After being dried, the area appears white and chalky. *Caries in pits and fissures:* Grade I: Area is dark by incident as well as transmitted light; the lesion is confined to a small dark line. Grade II: In addition to Grade I, a white zone can be seen along the margins of the fissure, which appears dark in transmitted light. Grade III: There is smallest perceptible break in the continuity of the enamel.
Radike AW (1968)	I. *Frank lesions (lesions without cavitation):* Visible presence of a cavitation. A. Cavitation as a result of discontinuity of the enamel surface caused by loss of tooth surfaces. B. Cavitation as a result of the caries process should be distinguished from fractures and smooth lesions or erosion and abrasion. II. *Lesions not showing cavitation:* The criteria for detection of these lesions are categorized as: A. Detection of pit and fissure lesions of the occlusal, facial, and lingual surfaces. 1. Area is carious when the explorer 'catches' or resists removal after insertion into a pit or fissure with moderate to firm pressure and when accompanied by one or more of the following signs of caries: a. A softness at the base of the area b. Opacity adjacent to the pit or fissure as evidence of undermining or demineralization c. Softened enamel adjacent to the pit or fissure which may be scraped away with the explorer 2. Area is carious, if there is loss of the normal translucency of the enamel, adjacent to a pit, which is in contrast to the surrounding tooth structure. This condition is considered as an evidence of undermining. In some cases, the explorer may not catch or penetrate the pit. B. Detection of lesions on smooth surfaces (facial and lingual) 1. Area is carious, if surface is etched or if there is a white spot as evidence of subsurface demineralization, and if the area is found to be soft by: a. Penetration with explorer b. Enamel can be scraped away with explorer 2. Area is sound when there is apparent evidence of demineralization (etching or white spots) but no evidence of softness. C. Detection of lesions on proximal surfaces: The procedures used for diagnosis of proximal surfaces varied considerably. Some examiners depend largely upon visual-tactile methods; whereas others on radiographs; a few have used the combination of these procedures. The inference is: 1. For area exposed to direct visual and tactile examination, these are diagnosed as under 'B' above for smooth areas. 2. For hidden areas not exposed to direct visual-tactile examinations: a. *Visual examination:* If the marginal ridge shows an opacity as evidence of undermined enamel, the proximal surface is carious. b. *Tactile examination:* Any discontinuity of the enamel in which an explorer will enter is carious, if it also shows other evidence of decay, such as softness, shadow by transillumination, or loss of translucency. c. *Radiography:* Any definite radiolucency indicating a break in the continuity of the enamel surface is carious. d. *Transillumination:* A loss of translucency producing a characteristic shadow in a calculus-free and stain-free proximal surface is adequate evidence of caries. It is used mostly for anterior teeth.
Moller IJ and Paulsen S (1973)	A. Pits and fissures a. Discoloration not definite 'sticking' b. Sticking with or without discoloration; no dentin involvement

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Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective *(Contd...)*

	c. Definite cavity with dentin involvement d. Probable pulp involvement B. Vestibular and lingual smooth surfaces a. White opaque area with loss of luster; no loss of substance b. Discontinuity in the enamel; no dentin involvement c. Dentin involvement d. Probable pulp involvement C. Proximal surfaces (diagnosed by means of radiographs)
Murray J and Shaw L (1975)	C1: A pit, fissure of smooth surface in which there is discoloration in both incident and transmitted light, with minute discontinuity in the enamel surface which does not allow definite 'sticking' of the probe on gentle pressure C2: A cavity in a pit, fissure or a smooth surface in which a probe sticks with gentle pressure and requires a definite pull to be removed C3: A large open cavity, probably with pulpal involvement
Howat AP (1981)	A. Visual method I. Pits and fissures V1: Visibly intact enamel with one or both of the following; dark line at base of fissure and/or white discoloration on the walls V2: Breakdown of walls of fissure with either visible open cavity or shadow or opacity beneath the enamel (evidence of undermining), less than 1.5 mm across the fissure V3: Breakdown of walls of fissure with either a visible open cavity, or shadow or opacity beneath the enamel (evidence of undermining), greater than 1.5 mm across the fissure II. Free smooth surfaces V4: White chalky opaque lesion, but intact surface V5: Break in enamel with cavity less than 0.5 mm in diameter V6: Cavity greater than 0.5 mm in diameter III. Mesiodistal surfaces: V7: Shadow by transillumination, or loss of translucency V8: Observable cavity B. Tactile method I. Pits and fissures T1: Sticky on probing; the probe requires a definite pull to be removed T2: The explorer 'catches' or resists removal with moderate to firm pressure, and adjacent softened enamel may be scraped away with the explorer. T3: The explorer 'catches' or resists removal with moderate to firm pressure and is accompanied by softness at the base of the area. II. Free smooth surfaces T4: A surface discontinuity, including defective (soft or brittle) enamel, tip of probe sticking superficially T5: Cavity with a detectably softened floor T6: Cavity detected by probing; probe sticking superficially III. Mesiodistal T7: Cavity detected by probing, with a detectable softened floor C. Visual-tactile method I. Pits and fissures VT1: Fissure showing discoloration with incident as well as transmitted light and no definite stickiness VT2: The explorer 'catches' or resists removal in a pit or fissure with moderate to firm pressure, and is accompanied by opacity adjacent to the pit or fissure (evidence of undermining or demineralization). VT3: Visible cavity, probe penetrates dentin II. Free smooth surfaces VT4: White spots with loss of luster with hard intact surface VT5: White spots with softness detected by penetration and scraping away of enamel by an explorer VT6: Break in enamel in a chalky or brown-stained area of diminished translucency less than 0.5 mm in diameter, with soft floor VT7: Definite cavity with soft floor, diameter greater than 0.5 mm.

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Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective *(Contd...)*

	III. Mesiodistal VT8: White or brown area of enamel seen by direct vision; dark area of enamel seen by transillumination, no visible cavity entered by the probe (mostly for anterior) VT9: If the explorer detects roughness, softening, or surface discontinuity in a chalky or brown-stained area of diminished translucency
WHO (1987)	When a lesion in a pit or fissure, or on a smooth tooth surface, has a detectably softened floor, undermined enamel, or softened wall is categorized as 'caries'. A tooth with a temporary filling should also be included in this category. On proximal surfaces, the explorer should enter the lesion; in case of doubt, the lesion should not be recorded as caries.
WHO (1997)	When a lesion in a pit or fissure, or on a smooth tooth surface, has a definite cavity, undermined enamel, or a detectably softened floor or wall is categorized as caries. A tooth with a temporary filling, or one which is sealed but also decayed, should also be included in this category. In cases where the crown has been destroyed by caries and only the tooth is left, the caries is presumed to have originated on the crown and is scored as crown caries only. The CPI probe should be used to confirm visual evidence of caries on the occlusal, buccal, and lingual surfaces. Doubtful cases are not recorded as caries.
Pitts NB and Fyfee HE (1988)	A. *Initial caries:* No detectable loss of tooth substance. For pits and fissures, there may be significant staining, discoloration, or rough spots in the enamel that do not catch the explorer, but where loss of substance cannot be positively diagnosed. For smooth surfaces, these may be white, opaque areas with loss of luster. B. *Enamel caries:* Loss of tooth substance in pits, fissures, or on smooth surfaces, but no softened floor or wall or undermined enamel. The texture of the material within the cavity may be chalky or crumbly, but there is no evidence that cavitation has penetrated the dentin. C. *Caries of dentin:* Detectably softened floor, undermined enamel, or a softened wall, or the loss has a temporary filling. On proximal surfaces, the explorer should definite enter a lesion. D. *Pulpal involvement:* Deep cavity with probable pulpal involvement. Pulp should not be probed.
Pitts NB, et al. (1997)	Code 1: Arrested dentinal decay: In case the surface has arrested lesion. Code 2: If the active caries is extended into dentin. Code 3: If the caries is extended into pulp.
Amarante, et al. (1998)	Occlusal caries Grade 1: Caries characterized by white or brown discoloration without tooth substance loss. No radiographic findings. Grade 2: Little substance loss with break in the enamel surface or discoloring; caries restricted to enamel in the X-ray. Grade 3: Moderate tooth substance loss and/or caries in the external one-third of the dentin radiographically Grade 4: Considerable tooth substance loss and/or caries in the middle one-third of the dentin radiographically Grade 5: Substantial tooth substance loss and/or caries in the internal one-third of the dentin radiographically Similar grades were used for buccal and lingual surfaces and secondary caries.
Ekstrand, et al. (1998)	0 = no or slight change in enamel translucency after prolonged air drying 1 = opacity (white); hardly visible on the wet surface, but distinctly visible after air drying 1a = opacity (brown); hardly visible on the wet surface, but distinctly visible after air drying 2 = opacity (white); distinctly visible without air drying 2a = opacity (brown); distinctly visible without air drying 3 = localized enamel breakdown in opaque or discolored enamel and/or grayish discoloration from the underlying dentin 4 = cavitation in opaque or discolored enamel, exposing the dentin beneath
Nyvad, et al. (1998)	0 Sound= Normal enamel translucency and texture (slight staining allowed in otherwise sound fissure).

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Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective *(Contd...)*

	1= Active caries (intact surface): Surface of enamel is whitish/yellowish opaque with loss of luster; feels rough when the tip of the probe is moved gently across the surface; generally covered with plaque. No clinically detectable loss of substance. Intact fissure morphology; lesion extending along the walls of the fissure.
	2 = Active caries (surface discontinuity): Same criteria as score 1. Localized surface defect (microcavity) in enamel only. No undermined enamel or softened floor detectable with the explorer.
	3 = Active caries (cavity): Enamel/dentin cavity easily visible with the naked eye; surface of the cavity feels soft or leathery on gentle probing. There may or may not be pulpal involvement.
	4 = Inactive caries (intact surface): Surface of enamel is whitish, brownish or black. Enamel may be shiny and feels hard and smooth when the tip of the probe is moved gently across the surface. No clinically detectable loss of substance. Intact fissure morphology; lesion extending along the walls of the fissure.
	5 = Inactive caries (surface discontinuity): Same criteria as score 4. Localized surface defect (microcavity) in enamel only. No undermined enamel or softened floor detectable with the explorer.
	6 = Inactive caries (cavity): Enamel/dentin cavity easily visible with the naked eye; surface of the cavity feels shiny and feels hard on gentle probing. No pulpal involvement.
	7 = Filling (sound surface): Same as score 0.
	8 = Filling with an active caries: Lesion may be cavitated or non-cavitated.
	9 = Filling with an inactive caries: Lesion may be cavitated or non-cavitated.
Fyfee, et al. (2000)	1. *White-spot lesion:* Visual examination indicates intact surface; no clinically detectable loss of tooth substance, with a white or cream-colored area of increased opacity, presumed to be caries. 2. *Brown-spot lesion:* Visual examination indicates intact surface; no clinically detectable loss of tooth substance, with a brown/black discoloration, presumed to be caries. 3. *Enamel cavity:* There is a lesion with demonstrable loss of surface but no visual, clinical evidence of the lesion penetrating dentin. 4. *Dentin lesion (uncavitated):* There is a caries lesion into dentin but no visible evidence of cavitation. 5. *Dentin cavity:* There is a carious cavity into dentin. 6. *Pulp involved:* There is a carious cavity that involves the pulp. 7. *Arrested dentinal decay:* There is arrested caries in dentin.
ICDAS, II	0 = Sound tooth surface: No evidence of caries after prolonged air drying (5 s).
(2009)	1 = First visual change in enamel: Opacity or discoloration (white or brown) is visible at the entrance of the pit or fissure after prolonged air drying, which is not or hardly seen on a wet surface.
	2 = Distinct visual change in enamel: Opacity or discoloration distinctly visible at the entrance of the pit and fissure when wet, lesion must still be visible when dry.
	3 = Localized enamel breakdown due to caries with no visible dentin or underlying shadow: Opacity or discoloration wider than the natural fissure/fossa when wet and after prolonged air drying.
	4 = Underlying dark shadow from dentin +/– localized enamel breakdown.
	5 = Distinct cavity with visible dentin: Visual evidence of demineralization and dentin exposed.
	6 = Extensive distinct cavity with visible dentin and more than half of the surface involved.
	Caries associated with restoration and sealants
	0 = Sound tooth surface with restoration and sealant.
	1 = First visual change in enamel.
	2 = Distinct visual change in enamel/dentin adjacent to restoration/sealant margin.
	3 = Carious defect of >0.5 mm, with signs of code-2.

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Table 7.1: Visual-tactile criteria for detection of caries: Authors' perspective *(Contd...)*

	4 = Marginal caries in enamel/dentin/cementum adjacent to restoration/sealant, with underlying dark shadow from dentin. 5 = Distinct cavity adjacent to enamel/dentin. 6 = Extensive distinct cavity with visible dentin.
Baurer, et al. (1988)	*Root caries:* Discrete, well-defined, and discolored cavitation on the root surface into which the explorer entered easily and displayed some resistance to withdrawal
Beighton, et al. (1993)	*Root caries:* The diagnosis of primary root caries was made by consideration of change in color, texture, and surface contour of the tooth. The color of each lesion was categorized by visual comparison with a standard guide of four shades (yellow, light brown, dark brown, and black) which was prepared from photographs of primary root caries lesions. The dimensions of each lesion, the occluso-gingival and mesiodistal lengths, and greatest loss of surface contour were measured with a periodontal probe marked with 1.0 mm intervals. The texture of the lesion was classified into three grades: Hard, leathery (assessed by means of a new Ash No.6 probe with moderate pressure but displaying resistance to its withdrawal), and soft lesions which were easily penetrated and displayed no resistance to withdrawal of the probe.
Rosen, et al. (1996)	*According to the location of the lesion:* Caries lesions transversing the cementoenamel junction and involving both the coronal and root portions of a tooth surface were classified as coronal caries. *Cavitation:* Lesions were classified as manifest caries as soon as the tooth structure has disintegrated as a cavity was formed. *Discoloration:* White or brown spot lesions on the crown and yellowish or brownish to black lesions on the roots were recorded as initial caries. *Surface structure:* The manifest lesions were classified as hard or soft and the initial lesions as smooth or rough.
Ekstrand, et al. (2008)	*Root caries* *Texture of the lesion* • Hard (0) • Leathery (2) • Soft (3) *Contour of the surface* • No cavitation or smooth cavity surroundings (1) • Cavitation with irregular border (2) *Distance from the lesion to the gingival margin* • More than 1.0 mm from the gingival margin (1) • Less than 1.0 mm from the gingival margin (2) *Color of the lesion* • Dark brown/black (1) • Light brown/yellowish (2)

Visual and tactile methods have been the hallmark of detection of carious lesions in the earlier studies. It is established as follows:

- The non-cavitated caries lesions are more prevalent than cavitated lesions in economically developed countries.
- Non-cavitated caries lesions are more likely to be restored compared with sound tooth surfaces.
- Non-cavitated lesions, especially on smooth tooth surfaces in young children, may serve as indicators of caries activity.
- Inclusion of non-cavitated lesions may provide a better understanding of the mechanism of action of fluoride, sealants, and other preventive agents.
- Inclusion of early signs of the caries process improves the precision of clinical trials of preventive agents.

Keeping in view these facts, emphasis is being given to diagnostic criteria, which can detect incipient carious lesions.

Disadvantages

- The visual tactile examination method, in all probability, may miss out an incipient lesion.
- The criterion for detection of caries is not standardized. The method of examination and criterion for diagnosis vary from individual-to-individual.
- Probing depth and pressure is controversial.

3. Radiographic Methods

The carious process leads to demineralization of the affected area of the tooth and that area appears radiolucent on the radiograph. The presence of radiolucency in the tooth substance is considered as caries. Radiographs have certain limitations; therefore, the combination of clinical examination along with the radiographic aid is considered mandatory for diagnosing caries. The mere presence of radiolucency may be misleading. The following radiographic methods are used in routine.

a. Conventional Radiography

Conventionally, two types of technique are employed:

i. Intraoral periapical radiography
ii. Bitewing radiography

Others modalities, like occlusal radiographs and panoramic radiographs are rarely employed in the detection of caries.

i. ***Periapical radiographs*** are useful for detecting changes in and around the tooth tissues, such as extent of caries, cervical margins of the restoration, alveolar crest height, lamina dura as well as the size of the pulp chamber. The paralleling technique is considered superior to bisecting technique for detecting caries in both anterior and posterior teeth (Fig. 7.1).

ii. ***Bitewing radiographs*** are important to detect incipient lesions at the contact areas. With this technique, six to eight teeth in one radiograph can be visualized. The technique can be used for anterior as well as posterior teeth. The film is available in different sizes to suit the area to be radiographed (Fig. 7.2).

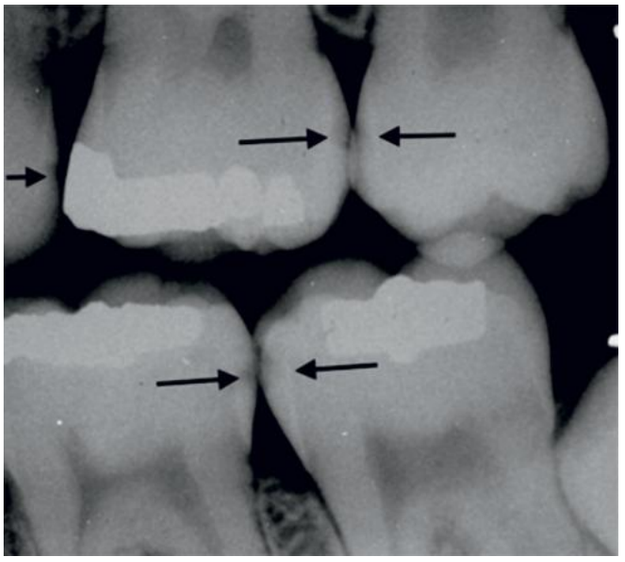


Fig. 7.2: Bitewing radiograph (arrow showing caries)

Posterior bitewing radiographs are preferably utilized to detect interproximal caries. Recurrent caries at the cervical margins is best observed in bitewing films, since the central ray is directed along the plane of the cervical areas.

Radiographic appearance of caries:

i. *Occlusal caries:* Radiographs are not effective until the occlusal caries involves dentin. Once in dentin, the radiographic image shows a broad based thin radiolucent zone in the dentin with little or no change apparent in the enamel (Fig. 7.3).
A significant manifestation of occlusal caries in dentin is the presence of a band of increased opacity at the base of carious lesion near the pulp



Fig. 7.1: Intraoral periapical radiograph (maxillary teeth)

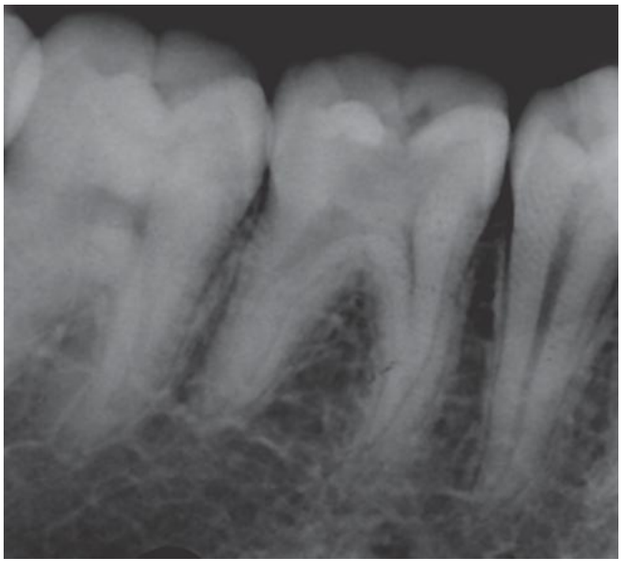


Fig. 7.3: Occlusal caries (mandibular first molar)

chamber. The white band represents the calcification within the primary dentin; such a band may not be evident in buccal caries.

Limitations of radiographic interpretation of occlusal caries are:

- Difficult to detect caries in enamel because of superimposition of adjacent enamel over the fissures.
- Lesions involving buccal grooves of molars are superimposed over the occlusal area and can simulate occlusal lesions.
- A thin radiolucency at the dentinoenamel junction in occlusal caries may be confusing with normal radiolucency in enamel and dentin.
- Difficult to distinguish between occlusal caries and the internal resorption.

ii. *Proximal caries:* A considerable loss of mineral content is mandatory before it becomes visible on a radiograph. The actual depth of the lesion is always deeper than may be seen radiographically (Fig. 7.4).

The white chalky appearance is the first evidence on the surface of enamel just below the contact point. The caries susceptible zone is 1.0 to 1.5 mm below the contact point, which is evident as a small radiolucent area below the contact area. With the advancing lesion, the radiographic image is like a diffuse triangle with the base at the surface of the tooth. Once the lesion crosses the dentinoenamel junction and invades into the dentin, the lesion appears as another triangle with the base at dentinoenamel junction.

iii. *Root caries:* Root lesions are usually having ill-defined saucer-like appearances and are progressive in nature. These are usually observed within 2.0 mm area of cementoenamel junction.

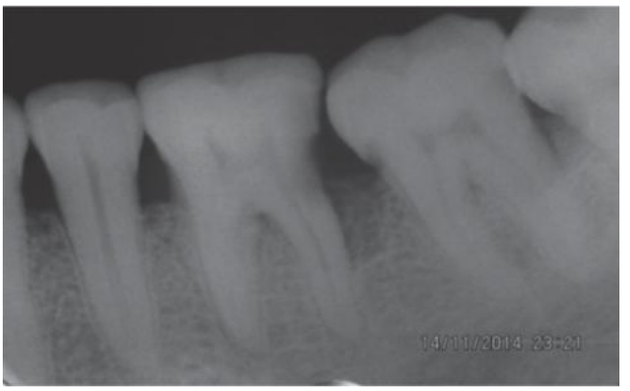


Fig. 7.5: Root caries (mandibular first molar)

An exposed root surface is always at an increased risk of developing caries (Fig. 7.5).

Diagnosis of root caries is not difficult, except in cases where the lesion is on interproximal surfaces (severe periodontal involvement lead to exposure of roots from all sides). Radiographic picture shows area of radiolucency in cementum alone or in cementum and dentin both.

iv. *Secondary/recurrent caries:* Caries that occurs adjacent to the restorations is known as secondary/recurrent caries (Fig. 7.6). Diagnosis of secondary caries is usually dependent on the clinical examination as the radiographs do not detect the lesion until only in advanced stages. The limitations in diagnosis of secondary caries are:

- Lesions between restoration and tooth surface cannot be visualized until reached to an advanced stage. (Discoloration of margins can be due to extrinsic stains or corrosion products.)
- Radiographs can diagnose the cervical portion of the interproximal lesions which is a prime site for secondary caries; however, radiolucency

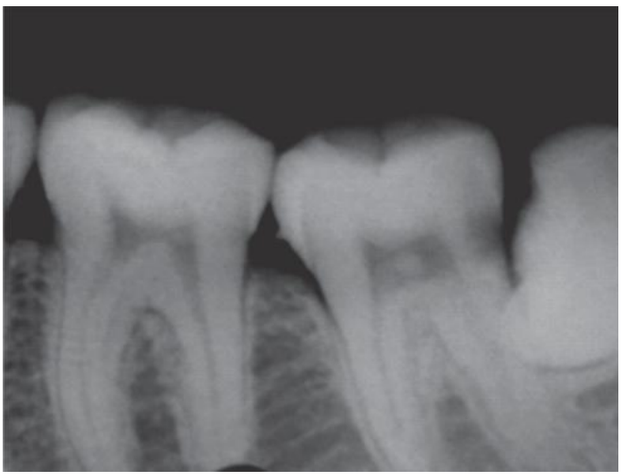


Fig. 7.4: Proximal caries (distal of mandibular first molar)



Fig. 7.6: Secondary caries (arrow)

under composite restorations may be deceptive because the bonding agents are also radiolucent.
- Difficult to differentiate between secondary caries and caries which has been left during restorative procedures (residual caries).

v. *Hidden caries:* Hidden (occult/covert) caries is a term used to describe a carious lesion, whereby the occlusal surfaces appear intact, and the radiographs show radiolucency in dentin. These caries develop through tiny enamel defects, progressing under seemingly intact tooth structure (Fig. 7.7). One of the reasons quoted is remineralization of the superficial enamel caries while the dentin is decaying underneath. Earlier authors were of the view that occlusal surface gets remineralized using fluorides and termed them as 'fluoride bombs' or 'fluoride syndrome'. They opined that fluoride encourages remineralization and slows down progress of the caries in the pit and fissure enamel, while the cavitation continues in dentin, and the lesion becomes masked by a relatively intact enamel surface. Later this hypothesis was challenged. Current opinion regarding the pathogenesis of occult lesions is focused on traditional mechanisms of caries development. It is presumed that a proportion of these lesions began their course as fissure caries, which might have progressed to hidden caries because of remineralization of occlusal surfaces.

Limitations of conventional radiography:

- Radiographs are two-dimensional image of a three-dimensional object.
- Overlapping of proximal contacts
- Lesion depth may appear increased or decreased due to change in angulation
- Difficulty in analyzing occlusal lesions

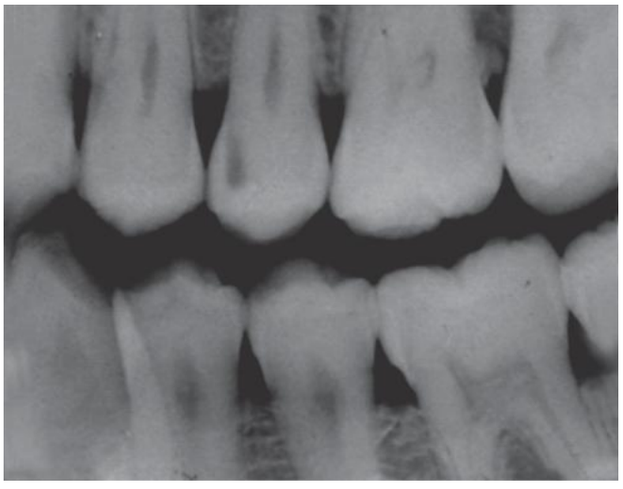


Fig. 7.7: Hidden caries (maxillary second premolar)

- Radiolucency, whether because of caries or resorption or any other defect is difficult to analyze.
- Demineralized area on the buccal/lingual surfaces may appear as proximal carious lesion.
- Fracture of one lingual cusp may appear as radiolucent proximal lesion.
- Tilt of maxillary lateral incisors appears as caries on the mesial side of the lateral incisors.
- Cervical burnout areas may mimic cervical caries.
- Conventional radiography may not correlate the relationship between depth of the radiographic lesion and the clinical cavitation.

b. *Xeroradiography*

Xeroradiography uses xerographic copying techniques to record images produced by X-radiations. The technique has an additional feature called 'edge enhancement' effect. Due to this effect, the tissues with subtle density differences are clearly visible.

In this technique the image is recorded on an aluminium plate coated with a layer of selenium particles. These selenium particles are given a uniform electrostatic charge and are stored in a unit called 'conditioner'. When X-rays are passed onto the film, it causes selective discharge of the particles. This forms the latent image and is converted to a positive image by a process called 'development' in the processor unit. The main characteristics of xeroradiographic technique are the ability to have both positive and negative prints together. When positive current is applied to the film, negative particles are attracted and when negative current is applied, positive particles are attracted.

Xeroradiography is twice as sensitive as conventional D-speed films.

Advantages

- Edge enhancement (two tissues of varying densities can easily be distinguished).
- One-third of X-radiation is required.
- Real image is produced.
- Reflected light is used.
- No need for chemical processing in dark room.
- Economical/cost-effective.

Disadvantages

- The electric charge over the film, causes discomfort to the patient. (Humid environment of oral cavity acts as a medium for flow of current.)
- Exposure time varies, as manufacturers do not indicate the exact thickness of the plate.
- Process of development cannot be delayed and is to be completed within 15 minutes.

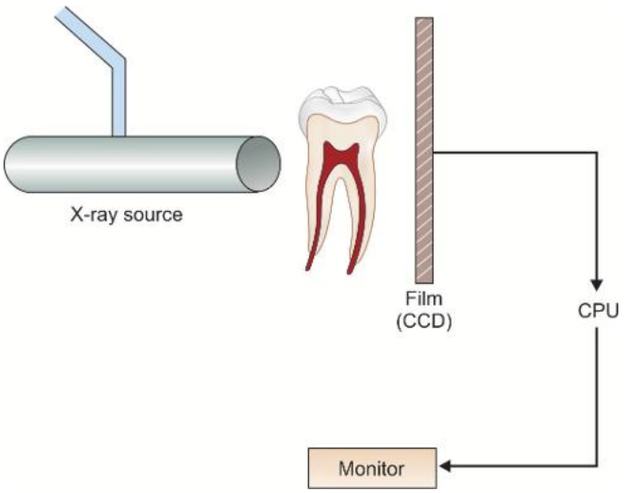


Fig. 7.8: Diagrammatic representation of digital imaging

c. Modified Radiographic Techniques

i. *Direct digital radiography:* Digital radiographic images may be obtained by digitization of a conventional film radiograph or by direct digital radiography [Radiovisiography (RVG)].

Digital images are acquired either directly (using a sensor or imaging plate replacing conventional film) or indirectly (by scanning and digitizing a film-captured image) (Fig. 7.8)

Direct digital imaging systems are divided into two types:

- Real time (cord system)
- Photostimulable phosphor imaging (cordless system)
- *Real time (cord systems):* These systems employ conventional X-ray generating equipment; however, conventional film is replaced by either a CCD (charge coupled device) or a CMOS (complementary metal oxide semiconductor) sensor, which is connected to the computer via a cable (or cord). The X-ray photons that reach the sensor are converted to light, by an intensifying or scintillation screen, which is picked by the CCD/CMOS and converted into an electrical charge, which once relayed to the computer, produces an almost instantaneous digital image on the monitor (hence the term real time).
- *Photostimulate phosphor imaging (cordless systems):* These systems employ reusable photo-stimulable phosphor imaging plates (PSPP) instead of film. The plates contain a layer of barium fluorohalide phosphor.

The phosphor layer absorbs and stores the X-ray energy that has not been attenuated by the patient. The image plate is then placed in a reader where it is scanned by a laser beam. The stored X-ray energy in the phosphor layer is released as light, which is detected by a photo-multiplier. From here, the information is relayed to the computer and displayed as a digital image on the monitor. The time taken to read the plate depends on the particular system being used, and on the size of the plate, but usually varies between 1 and 5 minutes.

Advantages

- Lower doses of radiation required as both types of digital image receptors are more efficient than conventional films.
- No need for conventional processing, thus avoiding all hazards associated with handling the processing solutions, including film faults.
- Easy storage and archiving of patient records.
- Easy transfer of images (teleradiology)
- Image enhancement and processing. Several image enhancement techniques are available, viz.
  - Inversion (reversal)
  - Alteration in contrast
  - Embossing
  - Magnification
  - Coloring various tissues

Disadvantages

- Long-term storage of the images; although CDs may be helpful.
- Digital image not secured especially for longer times.
- Intraoral placement of sensors may be difficult.
- Loss of image quality and resolution on the hard copy print.
- Image manipulation may mislead the inexperienced operator.
- Image can be tampered.

ii. *Computer image analysis:* A number of computer softwares are available, which may overcome the shortcomings of human eye, making the analysis easy. The observer variations in the interpretation of radiographs are well known. To avoid shortcomings of visual interpretation, softwares are being used for automated interpretation of digital radiographs. Automated analysis provides sensitive and objective observations, which may also permit the detection of small lesions that otherwise, may not be perceptible to naked eyes. Softwares are also available, which can quantify the extent of carious lesions.

Advantages

- Provides sensitive and objective observation, especially of smaller lesions, which otherwise may not be perceptible to naked eye.

- Lesion can be monitored.
- Small lesions can be quantified.

Disadvantages

- Exposure geometry needs be standardized.
- Specificity is less; however, sensitivity is better.
- Time consuming

iii. ***Subtraction radiography:*** It is a technique in which the 'structured noise' is reduced in order to increase the detectability of different tissues/objects on the radiograph. The structured noises are the images, which are not of diagnostic value and interfere in routine interpretation of radiographs. Subtraction images can be obtained from photographic and electronic methods.

Digitization is achieved by taking a picture of the radiograph using high quality video camera. This is fed to a computer-imaging device, termed as digitizer. Two standardized radiographs produced with identical exposure geometry are used. The first one is the 'reference image' and the subsequent images are for comparison. The reference image is displayed on the screen. Then the subsequent images are superimposed. The difference between the original and the subsequent images will show as dark bright areas, which can be interpreted readily.

Subtraction radiography is considered superior to conventional film radiography for detecting proximal and recurrent caries. It is also useful in detecting the progress of remineralization and demineralization patterns of dentinal caries. The assessment of alveolar bone height in determining the progression of periodontal disease has been one of the major uses of subtraction digital radiography (Fig. 7.9A to D).

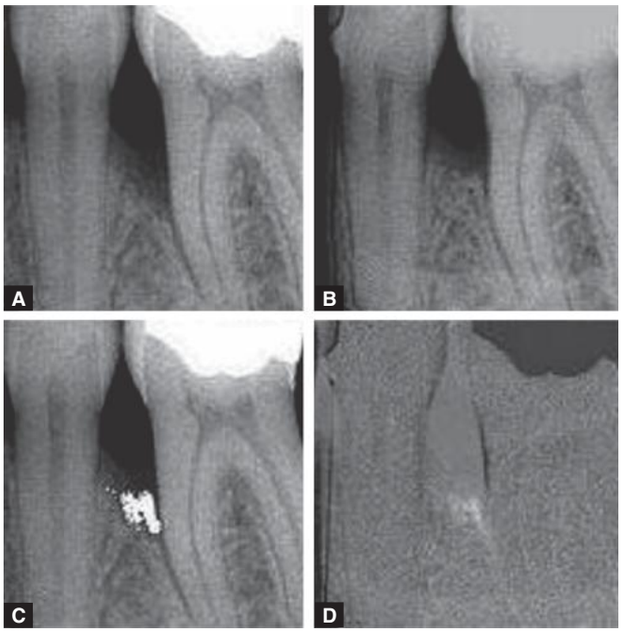


Fig. 7.9: Interdental bone formation seen in subtraction radiography

iv. ***Computed tomography (CT) scan:*** A computed tomography (CT) scan is a non-invasive diagnostic test that uses X-rays and a computer to create three-dimensional images of the object. This cross-sectional imaging technique provided an in-depth insight into the object scanned.

Computed tomography (CT) produces cross-sectional images of any part of the body. Each cross-sectional image represents a 'slice' of the area being imaged. These cross-sectional images are used for a variety of diagnostic purposes.

In a CT scan system, the X-ray source and the detector are situated at 180° from each other; their 360° movement around the patient, continuously detect information as the X-rays pass through the object. The computer further manipulates and integrates the acquired data and assigns numerical values based on the relative attenuation of the X-ray beam.

Refinements in detector technology lead to new CT scanners obtaining multiple slices in a single rotation. These scanners are called multislice CT or multidetector CT.

Local computed tomography (LCT) is another CT device introduced for caries detection.

The use of CT in caries diagnosis is limited because of high cost, excessive radiation, and not being user friendly.

v. ***Microcomputed tomography (micro-CT) scan:*** Non-destructive techniques of analyzing mineral changes in dental tissues are being tried. The tissue can be analyzed enabling evaluation of mineral loss/gain and also its kinetics.

The expressive potential of the micro-CT is for evaluation of mineral content in dental hard tissue. It also provides insights into the process of remineralization of early carious lesions.

The difference or change in mineral density is the basis of hard tissue research, vis-à-vis caries, and X-ray microtomography provides accurate approach as compared to conventional methods (Fig. 7.10).

vi. ***Cone beam computed tomography (CBCT):*** Cone beam CT imaging was introduced with the aim to provide three-dimensional data with lower absorbed doses than the conventional CT and micro-CT. Cone beam CT is widely used in dentistry, particularly in caries diagnosis and endodontic applications. The use of cone beam CT

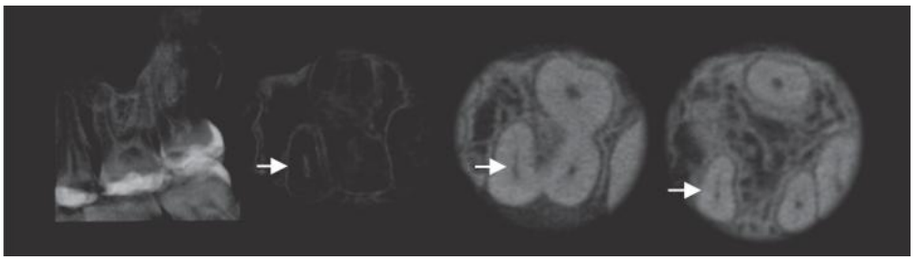


Fig. 7.10: Micro-CT images evaluating caries

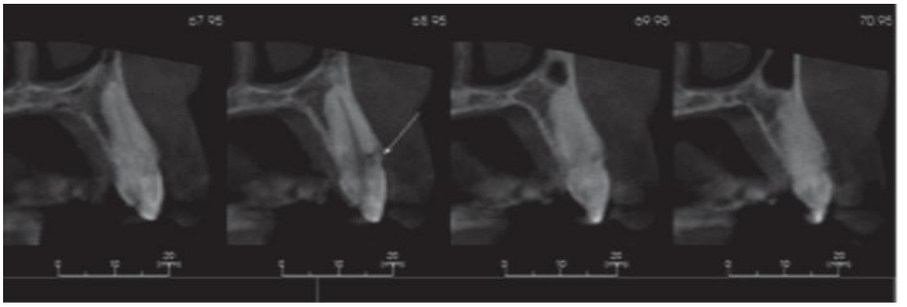


Fig. 7.11: CBCT image (Cross-sectional view): Maxillary left canine showing hidden buccal cervical lesion penetrating the pulp chamber and possibly ensuing the internal resorption

offers advantages over conventional tomography, viz. easier image acquisition, higher image accuracy, reduced artefacts, lower effective radiation doses, faster scan times and greater cost effectiveness. Rather than the fan-shaped beam emitted by conventional CT, cone beam CT emit a cone-shaped X-ray beam that can cover the region of interest (Fig. 7.11).

Spiral CT was developed with an aim to decrease radiations. The spiral CT requires a continuously rotating X-ray tube head. The patient is advanced continuously while the equipment rotates, in a spiral movement, around the patient. The investigation time has been shortened to only a few seconds with a radiation dose reduction up to 75%.

Whatever type of scanner is used, the level, plane and thicknesses (usually between 1.5 mm and 6.0 mm) of the slices to be imaged are selected and the X-ray tube head rotates around the patient, scanning the desired part of the body and producing the required number of slices (usually in the axial plane).

The sequence of events in image generation can be summarized as follows:

- As the tube head rotates around the patient, the detectors produce the attenuation or penetration profile of the body being examined.
- The computer calculates the absorption at points on a matrix formed by the intersection of all the generation profiles for that slice.
- Each point on the matrix is called a pixel and typical matrix sizes comprise either 512 × 512 or 1024 × 1024 pixels. The smaller the individual pixel, the greater the resolution of the final image.
- The area being imaged by each pixel has a definite volume, depending on the thickness of the tomographic slice, and is referred to as a voxel.
- Each voxel is given a CT number or Hounsfield unit between, +1000 and –1000, depending on the amount of absorption within that block of tissue.
- Each CT number is assigned a different degree of greyness, allowing a visual image to be constructed and displayed on the monitor.
- The patient moves along and adjacent sections are imaged.
- The selected images are photographed subsequently to produce the hard copy pictures with the rest of the images remaining on disc.

Disadvantages of CT scans

- Metallic objects, such as fillings may produce marked streak or star artifacts.
- Thin contiguous or overlapping slices may result in high dose.
- Inherent risks associated with contrast agents.
- Equipment is expensive.

The application of cone beam computed tomography (CBCT) in diagnosing dental caries is not established; however, CBCT has the definite

advantage of inducing the least amount of radiation hazard. CBCT has been showing similar results when compared to charge-coupled device (CCD) in detecting proximal caries.

vii. ***Magnetic resonance imaging (MRI):*** This technology is not available for clinical application. In the laboratory, the technique is capable of producing accurate three-dimensional reconstructions of teeth and also the caries lesions.
Recently developed MRI method, called Sweep Imaging with Fourier Transformation (SWIFT-MRI) has been claimed to overcome many limitations of the conventional MRI.
MRI is not suited for detection of caries.

4. Electric Resistance (Electrical Conductance and Impedance)

It is established that sound tooth enamel is a good electrical insulator due to its high inorganic content. Enamel demineralization results in increased porosity of enamel. Saliva fills these pores and forms conductive pathways for electric current. The electric conductivity is directly proportional to the amount of demineralization. Electric resistance is measuring the electrical conductivity through these pores.

Since saliva is a better electrical conductor than enamel tissue, the conductivity increases with increase of demineralization. Electrical resistance is also measured during controlled drying. By drying the tooth surface, the resistance is determined by the tooth structure, avoiding electrical conductance by saliva. The higher values indicate well-mineralized tissue; whereas, low values indicate demineralized tissue.

Factors affecting electrical measurements: The factors affecting electrical measurements of teeth are:
i. *Porosity of the tissues:* It is established that teeth undergo a maturation process after eruption; the process has a significant effect on porosity and hence electrical properties. There may be variations among individuals or within the same individual; usually within 1 to 1½ year of eruption, the electrical impedance characteristics vary.

Porosity in the context of electrical measurement means the pore depth, pore volume or the three-dimensional configuration of the pore; all can affect ionic migration within dental hard tissues. Maturation of teeth also causes changes in the porosity of the dental tissues.

ii. *Surface area of the electrode in contact with the tooth:* The surface area of the electrode in contact with the tooth is a significant factor in electrical measurements. The two modes of electrodes, 'site (point) specific' and 'surface-specific' produce different values for electrical properties; however, surface-specific mode is conventionally used.

iii. *Thickness of the tissues:* Thickness of the tooth tissues, especially the variation of enamel thickness, affects electrical measurements.

iv. *Hydration of the enamel:* Hydration of the enamel does affect electrical measurements. The electrical measurement devices are designed to operate with a specific air-flow; however, the teeth should not be dehydrated prior to measurement. Standardized use of contact medium is preferred to achieve uniformity.

v. *Temperature and concentration of ions in the fluid within the tooth:* It is demonstrated that the temperature of the teeth affects the electrical conductivity. The concentration of ions in the fluids within the tissues, particularly within enamel, is also critical to electrical measurement values.

a. Electronic Caries Meter (ECM)

The electronic caries meter employs single fixed-frequency alternating current (23 Hz) measuring the resistance of the tooth. The ECM probe is directly applied to the site, expressing compressed air from the tip of the probe.

It works on the principle of electrical conductivity. Sine waves of 400 Hz are utilized to measure currents for the caries meter. The electrical impedance is indicated by four-colored lights: Green, yellow, orange and red (Table 7.2).

The electrical conductivity reading on a scale from 0–9 indicates a change from sound tooth to an increased degree of demineralization (taken as caries).

Table 7.2: Color: Indications, impedance values and treatment options

Color	*Impedance values*	*Indication*	*Treatment options*
Green	600 k	Healthy tooth	No treatment required
Yellow	Between 250 and 600 k	Enamel caries	Observe for some time: Pit and fissure sealant
Orange	Between 15 and 250 k	Caries extending into dentin	Remove caries and restore
Red	Below 15 k	Caries approaching pulp; exposed to bacterial infection	Pulp removed; root canal treatment

b. *Caries Meter-L*

The caries meter-L utilizes 400 Hz current displaying four-colored lights, reflecting the status of the tooth (Fig. 7.12).

- Green—no caries
- Yellow—enamel caries
- Orange—dentine caries
- Red—pulpal involvement

The information obtained from the device remain insufficient for many reasons. Restricting and standardizing the flow of air to dry the tooth and the area of saline contact medium made the technique less favourable as compared to other machines.

c. *Vanguard Electronic Caries Detector*

The Vanguard electronic caries detector overcame the inconsistency in the direction and flow rate of air. The probe tip was placed centrally and coaxially within a stream of air so that when the probe was placed in the fissure, superficial saliva was removed while taking the reading. The conductance measurements were made between a specially designed probe tip and a hand-held connector.

The device is valuable in measuring early lesion, as the porosity produced in enamel is responsible for the drop in the resistance values; the dentin is markedly less resistant to the electric current. It is observed that a single reading at one visit may not be able to measure the activity of the lesion. When lesions are monitored longitudinally, it is possible to detect the change towards remineralization, demineralization or stabilization.

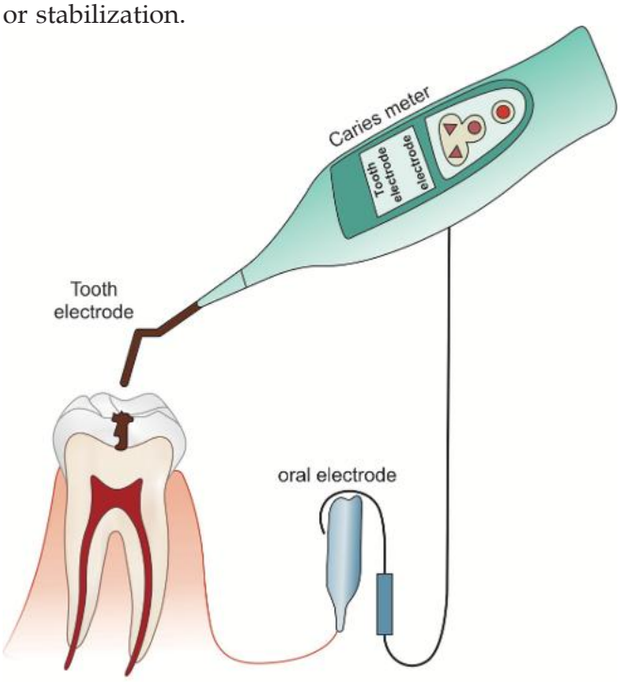


Fig. 7.12: Electronic caries detector

d. *Electronic Caries Meter (ECM) by LODE Diagnostics*

i. *Prototype ECM I:* The first prototype used a design close to the Vanguard machine. The conductance measurements were made between a tip placed on a handpiece and a hand-held reference electrode. Readings were also site-specific. The ECM reading appeared on a screen on the front of the device. It can be positioned in a range of about 1.00 to 13.00, representing increasing electrical conductance (Table 7.3). A higher reading meant more decay.

Table 7.3: Clinical interpretation of value ranges measured using the ECM prototype I

Range	*Clinical interpretation*
1.00 to 3.00	Sound enamel or early stages of caries
3.01 to 6.00	Caries up to the DEJ (enamel caries)
6.01 to 8.00	Dentinal caries
8.01 to 13.00	Deep dentinal caries

ii. *ECM II:* ECM II was battery-driven. The probe tip was in the center of an air tube of internal diameter 1.8 mm and had a diameter of 0.46 mm. A flow metre linked to the air supply was integrated so that the airflow around the probe tip could be adjusted. An audible beep indicated that the circuit was completed between the probe tip and the hand-held connector. A double beep indicated when the stable conductance reading was reached. A carious lesion confined to enamel depends on the degree of porosity and lesion depth.

ECM IIb has also been reported. It consisted of a probe tip connected to an alternating current supply. The digital display panel gave a resistance measurement on a scale 0–2 MW. A conducting medium was used to obtain a surface-specific reading. ECM was found to be the most valid and repeatable method used for enamel and dentin caries (Table 7.4).

Table 7.4: Cut-off points and their clinical implications as recommended for ECM IIb

Clinical implication	*Range*
Dentinal caries	0.000 to 0.390 MΩ
Enamel lesions	0.391 to 0.501 MΩ
Sound tissue	>0.501 M Ω

iii. *ECM III:* ECM III studied the effect of altering the protein content of enamel or etching. It was found that the ECM resistance readings decreased as the concentration and time of deproteinization (by sodium hypochlorite) increased. The acid etching reduced the resistance readings as well. The reduction of the solution to an acidic level (pH = 6) increases its efficacy

as an antimicrobial agent. Furthermore, at different concentrations, sodium hypochlorite lowers the pH value at the tissue level. These effects might explain the reduction of the ECM resistance as an acidic effect with the deproteinization one.

It was used to monitor lesions demineralization. It was found that the integrated value was inversely related to lesion depth and mineral loss in both enamel and root samples. It was also inversely related to the time to demineralization. Root caries was tested using the ECM in various ways. Soft dentinal lesions and dark brown lesions had a lower mean value than leathery lesions and light brown lesions, respectively.

iv. ***ECM IV:*** ECM IV used AC power supply by an adapter to 50 Hz–15 volt AC. It worked on a low frequency of 23 Hz and a current <0.3 mA. Air supply was through a coupling to the air rotor lead and activated automatically once there was electrical contact between the measuring and reference electrodes. The probe tip was situated in the center of an air tube. The flow rate was set to 5 L/min and this could be changed to 7.5 L/min when using the Vanguard measurement method.

The readings appeared by three procedures:

- *Position 0* used the 'Vanguard procedure' (Vanguard machine used previously). Once the circuit was closed, the start of the drying was indicated by an audible signal (one beep). The measuring time was determined by the behavior of the lesion during the drying period. When the resistance change was within a stable range for 3 seconds, drying stopped (a double acoustic beep was heard). Two values appeared: The first was whole numbers ordinal conductance scale (0–9), which was inversely related to the resistance and indicated increasing degrees of demineralization. The second was the measured resistance value (end value). The recommended cut-off values are presented in Table 7.5.

Table 7.5: Cut-off points and their clinical implications as recommended for ECM IV (Vanguard values)

Clinical implication	*Range*	*Cut-off points*
Sound enamel	0 to 2	>1 G to 10.0 MΩ
Initial caries	1 to 3	17.0 M to 4.00 MΩ
Caries up to DEJ	3 to 7	6.00 M to 1.50 MΩ
Caries passed DEJ	6 to 8	2.00 M to 1.00 MΩ
Deep dentin caries	7 to 9	1.50 M to 0.20 MΩ

- *Position 1* used the 'continuous reading' procedure and did not apply air-drying.
- *Position 2* used the 'Standard ECM Scale' procedure. The total drying and measuring time was fixed at 5 seconds.

e. Electric Impedance Tomography

Electrical impedance tomography (EIT) is based on the principle of electrical impedance spectroscopy (ECS). Unlike caries meter, which uses fixed frequency, EIT uses a range of frequencies and provides information on capacitance and impedance also.

f. Alternating Current Impedance Spectroscopy (CarieScan)

The alternating current impedance spectroscopy is used to quantify caries at an early stage. This device involves passing of an insensitive level of electrical current through the tooth to identify the presence and location of the decay. The CarieScan is not affected by optical factors, such as staining or discoloration of the tooth; it provides a qualitative value based on the disease state rather than the optical properties of the tooth. The device is indicated for the detection, diagnosis, and monitoring of primary dental caries on occlusal and accessible smooth surfaces, which are not clearly visible to the human eye. It cannot be used to assess secondary caries, the integrity of a restoration, root caries, and the depth of excavation.

Procedure

This device uses disposable sensors for single use and non-disposable sensors for routine use. The sensor contacts the tooth surface being examined, a disposable metal clip is placed over the corner of the patient's lip, which connects to the CarieScan to complete the circuit. During measurement, a green color display indicates sound tooth tissue, while a red color indicates deep caries requiring operative, and a yellow color associated with a range of numerical figures from 1 to 99 depict varying severity caries, which require only preventive care.

A modified version of CarieScan is the CarieScan-Plus, which is a wireless control system allowing data to be automatically captured, filed, and recalled electronically on a by-tooth, by-surface, by-date basis for dental health monitoring. This enhances communication with the patient as an aid to preventive motivation and caries control. Various studies comparing CarieScan with visual examination, bitewing radiograph, and DIAGNOdent reported CarieScan to have superior sensitivity and specificity (both 92.5%) over other methods.

5. Optical Detection Methods

It is recognized that the conventional methods are effective, only if at least one-third thickness of enamel

is involved. These methods rely on subjective interpretation and are insensitive to incipient caries.

The initial effects of the caries process (initial demineralization changes) results in an alteration in the optical properties of the affected dental tissues. Caries detection based on changes in a specific optical property are referred to as 'optical detection methods'. The methods are based on the measurement of a physical signal, derived from the interaction of light with dental hard tissue.

Caries process is the result of tilt of balance between the loss of mineral contents (demineralization) and the diffusion of mineral into the tooth (remineralization). The increase in demineralization is the start of carious process. The scattering of incident light is increased due to the structural change and appears as 'white spot'. The caries process leads to distinct optical changes that can be measured and quantified. Most of these methods use light in the visible and near-infrared region.

Let us understand the basic phenomena of optical changes.

i. *Scattering:* Scattering is the process of re-distribution of light in different direction (change of direction of photon without loss of energy).
ii. *Absorption:* Absorption is defined as the process by which photons are stopped by an object and the wave energy is taken in by the same object. The energy lost is mostly converted into heat or into another wave, which has less energy and hence, longer wavelengths.
iii. *Fluorescence:* Fluorescence is a phenomenon where the light is absorbed in a specific wavelength and then emitted in a higher wavelength. The energy absorbed is released by emission of light at a longer wavelength. The pattern of light absorption and remission in the dental tissues varies according to the excited wavelength.
iv. *Transillumination:* The light passing through the tooth tissue can be used to analyze the carious lesions. Wavelengths in the visible range (400–700 nm) are limited by strong light scattering, making it difficult to image the tooth structure.

a. Optical Caries Monitor

It is a non-invasive method based on the scattering of light by enamel crystals in relation to their surrounding environment. The scattering of light is quantified in enamel and also during the carious process. The clinical applications of optical caries monitor are established (preferred for smooth surface carious lesions).

The monitor consists of a light source, reference unit and a detection tip. The light is transported through a fiber bundle to the tip, which is placed against the tooth surface. The reflected light is collected and measured by the photo-diode of the different fibers in the tip.

Advantages
- Facilitates quantification
- Correlates with methods established for determination of mineral loss

Disadvantage
- Useful only for caries lesions on smooth surfaces.

b. Fiberoptic Transillumination (FOTI)

It is accepted that there are different indices of light transmission for decayed and sound tooth structures. Illumination is delivered by means of fiberoptics from the light source to the tooth structure. The resultant changes in light distribution (light traverses the tooth) are recorded as an image. Since decayed tooth has a lower index of light transmission than the sound tooth structure, the decay areas exhibit as a darkened shadow.

Fiberoptic transillumination is mainly utilized for the detection of proximal caries. More so, it has been established to be more effective in the anterior region. The use in the posterior region is associated with some difficulty. In posterior teeth, the light probe is positioned below the cervical margin of the tooth; light may not traverse properly (Fig. 7.13A, B).

Advantages
- Lesions, which remain undiagnosed radiographically, can be diagnosed easily.
- Simple and comfortable for the patients.
- No hazards of radiations.
- Less time consuming.

Disadvantages
- Record keeping is difficult.
- Subject to intra- and inter-observer variations.
- Difficult to place the light probe in posterior regions.

c. Digital Imaging Fiberoptic Transillumination (DIFOTI)

Digital imaging fiberoptic transillumination (DIFOTI) method was developed in an attempt to overcome the shortcomings of FOTI by combining FOTI and a digital camera using charge-coupled device (CCD). Images so captured are transferred to a computer for analysis. The CCD projects instantaneous images, which can be compared for clinical changes between several images of the same tooth over a given period of time. In addition, illumination and other conditions that may affect the quality of the image can be easily controlled.

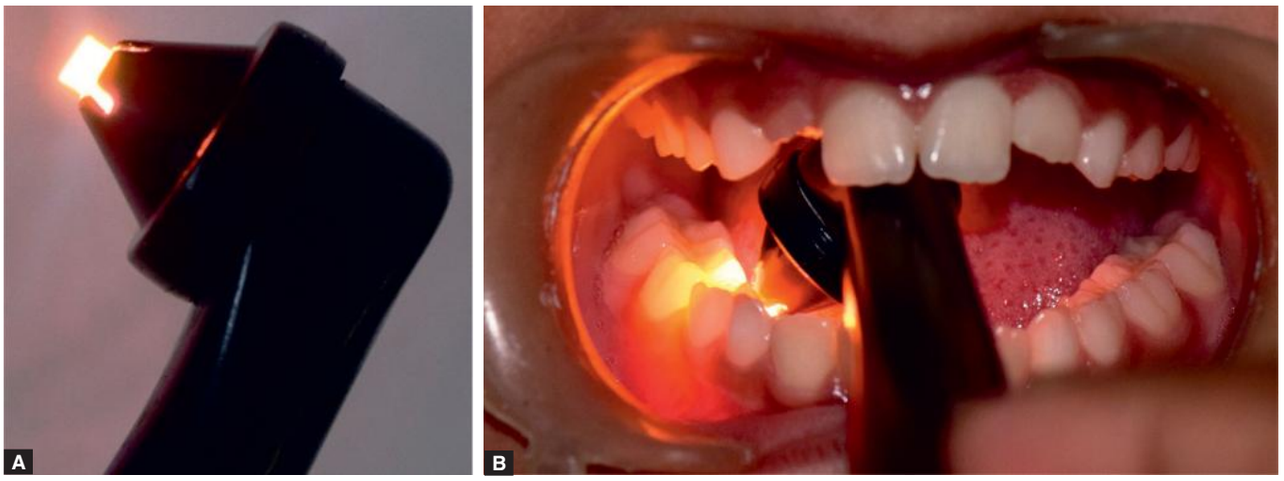


Fig. 7.13: Fiberoptic transillumination. **(A)** Fiberoptic tip; **(B)** Fiberoptic tip positioned at the cervical margin of posterior tooth

DIFOTI is being successfully used for the detection of incipient and recurrent caries. It enables the operator to confirm the presence of caries that cannot be seen radiographically, or felt by visual-tactile means.

It is important to note that DIFOTI images cannot indicate the depth of lesion penetration. The device is highly sensitive; dark areas in DIFOTI images may sometimes be due to stains or calculi on tooth surface. It is suggested that prophylaxis should be carried out prior to the use of the device in order to increase the specificity.

DIFOTI is twice sensitive in detecting proximal lesions and three times sensitive in the detection of occlusal lesions. For buccal-lingual lesions, the sensitivity was 10 times that of conventional radiographs.

Advantage

- Overcomes the disadvantages of FOTI

Disadvantage

- Do not indicate the depth of the lesion

d. Ultraviolet Illumination

Ultraviolet light has been used to discriminate the carious and the sound tissue of the tooth. The fluorescence (UV illumination) of tooth tissue is decreased in areas of less mineral content, such as in caries lesions.

The caries lesion appears as a dark spot against a fluorescent background. The fluorescent effects have been used as a method for the detection of dental caries in smooth surfaces. This method was more sensitive than visual/tactile methods. However, protective measures against ultraviolet radiation were required both for patient and the examiner. The specificity was also a problem in distinguishing between a caries lesion and other hypomineralized areas.

Advantage

- More sensitive than visual-tactile method

Disadvantage

- The carious lesion and the other areas of hypomineralization appear same.

e. Transillumination with Near-Infrared Light (TI-NIR)

The polarized light in the near-infrared range (1310 nm) gets depolarized by demineralized enamel and dentin. This rapid depolarization of polarized light by dental caries (demineralization) provides high contrast for imaging carious lesions (the dental tissues are highly birefringent; demineralization results in increase in light scattering and changes in birefringent).

The image can be captured, saved, and stored in digital format by use of transillumination with near infrared light. This is a promising imaging technique for detecting the presence of caries and also the extent of its severity. The method is non-destructive, non-ionizing, and reportedly more sensitive to detect early demineralization than dental X-rays.

Detection of dental caries by transillumination is based on the fact that increased mineral loss in enamel leads to a twofold increase in scattering coefficient. Caries thus appear as dark regions, since less light reaches the detector. When light illuminates the tooth, the strong scattering effect in the enamel caries lesions results in less transparency. The decreased light transmission associated with the lesion can be detected when compared to that of the surrounding sound tissue.

The transillumination method offers the advantage of allowing for repeated projection to overcome the

limitations of radiographs. The location of the caries lesion and how the resolution differs when the resultant image has to traverse a thick part versus a thinner part of the tooth is also important. The ratio between the contrasts of images captured from both sides of the tooth can estimate the precise location of the caries lesions.

Advantages
- Less amount of back scattering
- Can be easily differentiated from stains, pigmentation, and hypomineralisation (fluorosis)

Disadvantage
- May damage pulp

f. *Laser-induced Fluorescence*

A monochromatic light is used at 350 nm, 410 nm and 530 nm on carious and sound surfaces. In the carious lesions, the emission spectra shifts to more than 540 nm (red range of the electromagnetic spectrum). The largest difference between the carious and sound spectra is found at 600 nm. It is observed that when the enamel is illuminated with light in the blue-green range, the observable fluorescence occurs in the green-yellow range.

The tooth is illuminated with a broad beam of argon-ion laser (blue green light of 488 nm wavelengths) and the fluorescence observed in the yellow (540 nm) range. This fluorescence of enamel occurring in the yellow range (540 nm) is observed through a yellow high pass filter to exclude the tissue scattering blue light. Demineralized areas appear dark than the healthy tissue. Fluorescent dyes, such as Fluorol, sodium fluorescein, are also used to differentiate the areas of different mineral content. Demineralized tissues absorb dyes and fluoresce strongly. This is referred to as 'dye enhanced laser fluorescence'. A quantified version of laser fluorescence in the form of micro-camera has been developed to capture the real image. A computer screen displays the real images of the teeth under examination.

Advantages
- Convenient and relatively fast.
- Incipient carious lesions (minute mineral loss) can be detected
- Suitable for quantifying secondary caries (mineral loss around restorations).

Disadvantage
- Tooth surfaces should be absolute clean and dry.

Devices (with camera) using fluorescence principles

i. ***Quantitative light-induced fluorescence (QLF):*** The quantitative light-induced fluorescence (QLF) is based on the principle that the autofluorescence differs with the change in the mineral content of the dental hard tissue. Increased porosity due to demineralization results in loss of its natural fluorescence.

The QLF device uses a 50-watt xenon arc lamp and an optical filter producing blue light (290–450 nm wavelength), which is carried to the tooth. The fluorescence images are filtered and captured by a colored CCD camera. (The device consists of intraoral color CCD camera, a computer and the custom software to capture and analyze the images.)

When a lesion is present on the tooth surface an increase in scattering is observed relative to the surrounding normal tissues. The contrast is increased whereby the lesion appears as dark on the light green background of the tooth surface.

The fluorescence emitted is directly related to the mineral content of enamel. The change in enamel fluorescence is detected and measured.

Advantages
- Useful in detecting incipient lesions, secondary and root caries.
- Useful in detecting demineralization under orthodontic appliances/brackets, etc.
- Monitor extent of demineralization and remineralization.
- Provide quantitative parameters (lesion depth, area/volume)

Disadvantages
- Can't differentiate between decay, hypoplasia or unusual anatomic features.
- Potential for operator bias, position of camera may vary.
- Inability to detect/monitor interproximal lesions.
- May be influenced by stains, fluorosis/hypomineralization, etc.

ii. ***Light-induced fluorescence evaluation, diagnosis and treatment (LIFEDT) concept:*** The concept is based on the difference of fluorescence of dental tissues. The green fluorescence indicates healthy tissues; black green indicates infected dentin; bright red indicates the margin of infected/affected dentin; and, acid green fluorescence indicates sound dentin. These different fluorescence signals aid in caries detection. The principle is linked to the working of fluorescence based cameras.

Procedure
- Occlusal surfaces of tooth are cleaned.
- Can be observed in daylight or in a variety of modified forms.

- Reflected light from carious areas is noted and compared to healthy areas.
- System improves overall visual inspections.

The SoproLife camera (works on the principle of LIFEDT) has been launched to aid caries detection. The camera captures the images in three different modes that is, daylight, diagnosis and treatment mode. Capturing in the daylight provides a white light image with a magnification of more than 50 times than the tooth surface. The other two modes of the camera work on the principle of autofluorescence. In the diagnostic mode, the camera uses a visible blue light frequency (wavelength 450 nm) to illuminate the surface of the teeth, and provides an anatomic image overlay of the green fluorescence image on the 'white light' image. The green fluorescence is considered as an indicator of normal dental tissues; while carious lesions could be detected by variation in the auto-fluorescence of its tissues in relation to a healthy area of the same tooth. The third mode is the treatment mode, and the red fluorescence captured in this mode is considered as an indicator to differentiate between infected and affected dentin.

iii. ***New cameras:*** A new camera, the Soprococ, provides three modes: viz. daylight, caries and periodontal mode. VistaCam and vistaCam CL-IX work on the same principles with minor modifications.

Another camera based on light-induced fluorescence phenomenon is VistaProof intraoral camera. With this device, different areas of dental surfaces can be seen (green fluorescence—sound tissue; red fluorescence—carious tissue).

Devices (without cameras) using fluorescence principles

i. ***DIAGNOdent:*** DIAGNOdent follows principle of fluorescence without the use of a camera.

DIAGNOdent recognizes changes affecting the tooth, such as early demineralization and initial caries lesions. The gadget is also useful in determining the amount of decalcification in different areas of the same tooth. Even the residual caries can be identified (Fig. 7.14).

Procedure

The tooth/teeth to be assessed should be thoroughly cleaned and dried. After enamel is dried, light-scattering is increased, and a lower fluorescence can be measured. However, the performance (specificity and sensitivity) of the DIAGNOdent on dry tooth surfaces is not significantly different from that on moist surfaces. The fluorescence of a sound surface of the tooth is measured, which is considered as the baseline



Fig. 7.14: Conventional DIAGNOdent unit

value. This value is then subtracted electronically from the fluorescence of the site to be measured. Two different tips are used, one for occlusal site and the other for smooth surfaces. To achieve the maximum extension of caries on occlusal surfaces, one must tilt the instrument around the measuring site. This ensures that the tip picks up fluorescence from the slopes of the fissure walls, where the carious process often begins.

This system has a range of –9 and 99 with (–9 to –0) being the value of normal tooth. Initiation of caries is from 0–9; caries of enamel is from 10–17; and rest is for dentinal caries depending upon the depth. A new device, DIAGNOdent pen, has also been introduced (Fig. 7.15A, B).

Advantages

- 90% success rate.
- 0.92 sensitivity (more than electronic caries monitor).
- High reproducibility and reliability.
- Easy to use.
- Readily transportable
- Non-invasive and painless.
- No operator bias.

Disadvantages

- May induce false readings in the presence of plaque and debris.
- Readings do not relate exactly to the amount of dentinal decay; presence of micro-organism also matters.
- Cannot be used for recurrent caries.
- Staining, etc. can interfere with the results.

g. *Visible Luminescent Spectroscopy*

The visible emission spectra and the fluorescent lifetimes for decayed and non-decayed regions of teeth differ. Quasi-monochromatic light from a tungsten source dispersed with a grating mono-chromator was focused on the teeth. This is a non-

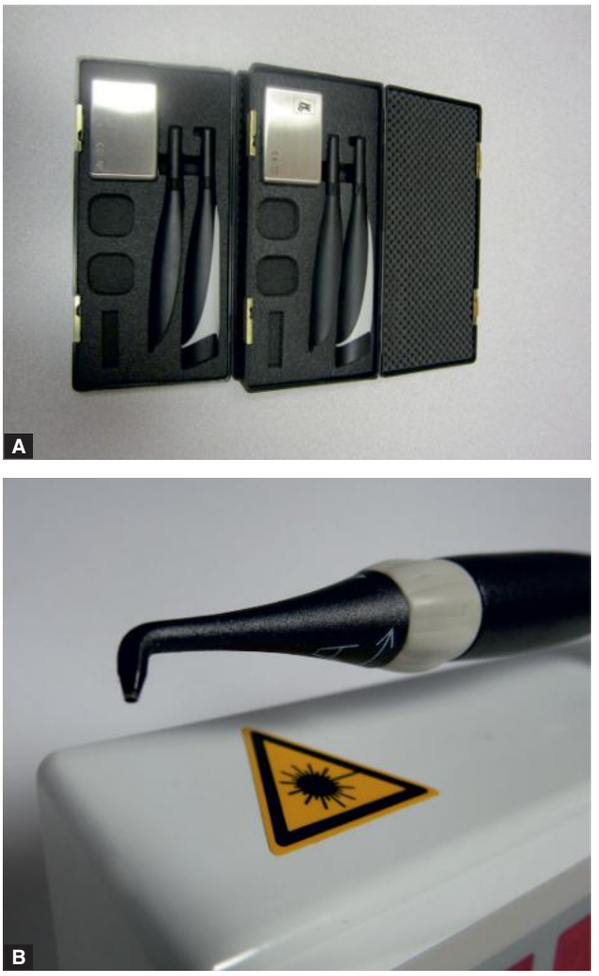


Fig. 7.15: DIAGNOdent Pen

radiological, non-invasive clinical method to detect dental caries.

h. *Terahertz Imaging (TI)*

Terahertz waves (frequency = 10^{12} Hz: wavelength 30 nm) are being used in caries diagnosis. This waveform provides reasonable resolution and also prevents loss of signal due to scattering. It was recognized earlier that photoconductive emitters could be used to generate and coherent picoseconds (10^{12} s) pulse, which would emit electromagnetic waves with a frequency in the terahertz range.

For an image to be obtained, the object is placed in the path of the terahertz beam; alternatively, the terahertz beam can be scanned over the surface of an object. Terahertz images have also been recorded using a CCD camera. A longitudinal image of human tooth demonstrated the outline of enamel–dentin junction as well as the dentin–pulp interface. Longitudinal sections further demonstrated increased terahertz absorption by early occlusal caries; also ability to discriminate dental caries from idiopathic enamel hypomineralization.

Advantages

- Relative transparency of human tissue to terahertz rays.
- Low powers used for imaging (~1 μW).
- Use of non-ionizing radiation
- No alteration of electrical charge of the oral tissues
- No adverse thermal effects
- Low signal-to-noise ratio (facilitates clear imaging).

Disadvantages

- Complexity of the laser source
- Precise manipulation mandatory
- Care is required in image interpretation, since terahertz waves are strongly absorbed by water, a potential complication in the oral cavity.
- High cost

i. *Multiphoton Imaging (MPI)*

Caries detection systems, such as QLF, rely on the fluorescence signal observed when teeth are exposed to blue light (~ 488–514 nm). This causes sound tooth structure to fluoresce. Carious tooth tissue may also fluoresce, but the disruption in the tooth structure results in profound scattering, which may result in little or no fluorescence. Consequently, sound tooth structure fluoresces at >520 nm; whereas, carious tooth tissue appears dark.

The choice of a longer wavelength of light for imaging reduces the scattering, allowing the light to penetrate more deeply within the tooth. This is the basis of multiphoton imaging (infrared light used at 850 nm), which makes any image of the tooth clearer.

Multiphoton imaging is able to collect information from caries lesions up to 500 microns in depth; however, the technique has been performed only on extracted teeth.

Advantages

- Provides three-dimensional image.
- A non-invasive method.
- Low speed of laser power used is less toxic to pulp.
- Enhanced depth of penetration due to longer incident wavelength used.
- Possibility of three-dimensional measurement of mineral loss.

Disadvantage

- May harm tissues; however, ultrashort pulses can overcome this problem.

j. *Optical Coherence Tomography (OCT)*

Optical coherence tomography was developed for imaging transparent and semitransparent structures. Teeth tissues are considered semitransparent.

Optical coherence tomography is based upon the principle of interference of light. When a light beam is split into two and then recombined, interference produces a pattern, the intensity of which is determined by the level of light in each beam. The OCT system uses superluminescent diodes (SLDs) as a light source. These diodes produce light with a broad range of wavelengths, each of which will produce its own interference pattern. The intensity of the interference is a function of the scattering caused by the changes in tissue structure of the tooth. Variation in scattering measured in relation to depth from a single point on the tooth surface is called an 'A-scan'. Taking several A-scans along a line produces information from a 'slice' of tooth tissue, which is the tomogram. The movement along the line of A-scans is known as the 'B-scan', and, it takes from 30 to 60 seconds to acquire a 1 cm long B-scan. A-scan is produced when light from a suitable source passes through a beam splitter and divides it into two coherent beams of light. One beam is called the 'sample beam' and the other, the 'reference beam'. The sample beam penetrates the tooth and gets scattered according to the nature of the tissue. It is established that carious tissues scatter light to a greater extent than does sound tooth structure. Some of the sample beam will be scattered back toward the beam splitter. The reference beam travels to a movable mirror, where it is reflected back to the beam splitter. Here, it is re-combined with the backscattered sample beam.

The caries lesions have been analyzed; changes in signal are related to the degree of scattering and possibly the degree of mineralization. OCT is being used to assess the restoration–tooth interface for semi-transparent restorations. This may have implications for the non-invasive diagnosis of secondary caries.

Advantages

- Clear images can be achieved.
- Quantitatively monitor the mineral changes in a caries lesion.
- Can determine severity of the lesion

Disadvantages

- False-positive results may be achieved due to back scattering.
- Limited to *in vitro* studies.

k. *Polarized Raman Spectroscopy (PRS)*

Polarized Raman spectroscopy measures light scattering as in optical coherence tomography. OCT imaging in regions of hypocalcification can sometimes show increased light back-scattering, which could be misinterpreted as signs of early caries. To rule out such false-positive readings and increase the specificity, OCT with PRS are combined to obtain information for carious lesions. PRS provides details on the molecular composition (e.g. collagen in dentin vs. predominantly inorganic apatite in enamel) and molecular structure of cells and tissues.

Advantages

- A non-invasive spectroscopic method analyzing biochemistry and molecular structure of white spot lesions.
- No need for external dyes.

Disadvantages

- Factors in the oral environment, such as calculus, hypocalcification, and stain could lead to false-positive results.

l. *Midwest Caries ID (LED Technology)*

Midwest caries ID, developed on LED technology, detects differences of optical behavior of sound tooth and the tooth with changes in their mineral content. The Midwest caries ID uses infrared and red light emitting diodes (LEDs) and a fiberoptic to distribute light to the given area of the object. A second fiber-optic collects light from that area to a photodetector that measures returned collected light. This photodetector then transmits the signal to a microprocessor that compares signal levels with defined parameters. When the detection is positive, the processor deactivates the third green LED; pulses at a higher intensity than the red LED. When the detection is negative (healthy tooth area), the green LED is dominant (green illumination in healthy tooth structure and red illumination in caries or demineralization). A buzzer also beeps with different frequencies to indicate the intensity of demineralization detected.

Advantage

- Intensity of demineralization can be measured.

Disadvantages

- False-positive results in case of teeth with growth malformations, dark stains, hypermineralization, hypocalcification, dental fluorosis, etc.
- Light penetration is limited (up to 3.0 mm in proximal area).

m. *Frequency-Domain Photothermal Radiometry and Modulated Luminescence (PTR/LUM)*

The combined frequency-domain laser-induced photo-thermal radiometry and modulated luminescence (PTR/LUM) is the recent development in caries diagnosis.

The technique is based on the photothermal response of a medium, resulting from optical radiation absorption from a low intensity laser beam. This is followed by optical-to-thermal energy conversion leading to slight

rise in temperature. In PTR applications to dental hard tissue, optical and thermo-physical properties and depth information are obtained in two distinct modes: (i) From near surface distances (5–500 μm) controlled by the thermal diffusivity of enamel and the modulation frequency of the laser beam intensity, and (ii) through mid-infrared emissions from considerably deeper regions. The PTR signal consists of both surface and subsurface responses of dental tissue; i.e. why it can distinguish between caries, stains on tooth surface, and developmental white spots, unlike the fluorescence device, such as QLF.

Advantages

- It is a depth profilometric technique while other photonic based technologies are not.
- Sensitive to changes in both optical and thermal properties of the sample
- Has the highest signal dynamic range in detecting very early demineralization
- Reliable and non-invasive

Disadvantage

- Signal reception need be improved.

n. *Time-correlated Single-photon Counting Fluorescence Lifetime Imaging*

The fluorescence lifetime imaging microscopy (FLIM) has also been used to distinguish carious areas from the sound dental tissue. A 488 nm excitation wavelength is required to perform time-correlated single-photon counting fluorescence lifetime imaging (TCSPC-FLIM) of dental tissues.

Advantage

- Relatively safe as compared to two photon excitation method.

Disadvantage

- Time consuming.

The advantages and disadvantages of optical methods used for caries detection are summarized in Table 7.6.

Table 7.6: Advantages and disadvantages of Optical methods

Diagnostic method	*Advantages*	*Disadvantages*
Optical caries monitor	• Facilitates quantification • Correlates with methods established for determination of mineral loss	• Useful only for caries lesions on smooth surfaces.
Fiberoptic transillumination (FOTI)	• Non-ionizing • Gives instant images • Lesions, which remain undiagnosed radiographically, can be diagnosed easily. • Simple and comfortable for the patients. • No hazards of radiations. • Less time consuming.	• Record keeping is difficult. • Subject to intra- and inter-observer variations. • Difficult to place the light probe in posterior regions.
Digital imaging fiberoptic transillumination (DIFOTI)	• Overcomes the disadvantages of FOTI	• Do not indicate the depth of the lesion • Stains/calculus may create problems
Ultraviolet illumination	• More sensitive than visual-tactile method	• The carious lesion and the other areas of hypomineralization appear same.
Transillumination with near-infrared light (TI-NIR)	• Less amount of back scattering • Can be easily differentiated from stains, pigmentation, and hypomineralization (fluorosis)	• May damage pulp (not documented)
Laser-induced fluorescence	• Convenient and relatively fast. • Incipient carious lesions (minute mineral loss) can be detected • Suitable for quantifying secondary caries (mineral loss around restorations).	• Tooth surfaces should be absolute clean and dry.
Quantitative light-induced fluorescence (QLF)	• Useful in detecting incipient lesions, secondary and root caries. • Useful in detecting demineralization under orthodontic appliances/brackets, etc. • Monitor extent of demineralization and remineralization • Provide quantitative parameters (lesion depth, area/volume)	• Cannot differentiate between decay, hypoplasia or unusual anatomic features. • Potential for operator bias (position of camera may vary). • Inability to detect/monitor inter-proximal lesions. • May be influenced by stains, fluorosis/ hypomineralization, etc.

Contd...

Table 7.6: Advantages and disadvantages of Optical methods (*Contd...*)

Diagnostic method	*Advantages*	*Disadvantages*
DIAGNOdent	• 90% success rate. • 0.92 sensitivity (more than electronic caries monitor). • High reproducibility and reliability. • Easy to use. • Readily transportable • Non-invasive and painless. • No operator bias.	• May induce false readings in the presence of plaque and debris. • Readings do not relate exactly to the amount of dentinal decay; presence of micro-organism also matters. • Cannot be used for recurrent caries. • Staining, etc. may interfere with the results.
Terahertz imaging (TI)	• Relative transparency of human tissue to terahertz rays. • Low powers used for imaging (~ 1 μW). • Use of non-ionizing radiation. • No alteration of electrical charge of the oral tissues. • No adverse thermal effects • Low signal-to-noise ratio (facilitates clear imaging).	• Complexity of the laser source • Precise manipulation mandatory • Care is required in image interpretation, since terahertz waves are strongly absorbed by water, a potential complication in the oral cavity. • High cost
Multiphoton imaging (MPI)	• Able to collect information up to 500 microns in depth. • Provides 3-D image. • A non-invasive method. • Low risk of phototoxicity to the pulp (low average speed of laser power used.) • Enhanced depth of penetration due to longer incident wavelength used. • Possibility of quantifiable measurement of mineral loss.	• May harm tissues; however, ultrashort pulses can overcome this problem • Technique performed on extracted teeth only
Optical coherence tomography (OCT)	• Clear image can be achieved. • Quantitatively monitor the mineral changes in a caries lesion. • Can determine severity of the lesion.	• False-positive results may be achieved due to back scattering. • Limited to *in vitro* studies.
Polarized Raman spectroscopy (PRS)	• A non-invasive spectroscopic method analyzing biochemistry and molecular structure of white spot lesions. • No need for external dyes.	• Factors in the oral environment, such as calculus, hypocalcification, and stain could lead to false-positive results.
Midwest caries ID (LED technology)	• Intensity of demineralization can be measured	• False-positive results in case of teeth with growth malformations, dark stains, hypermineralization, hypocalcification, dental fluorosis, etc. • Light penetration is limited (up to 3.0 mm in proximal area).
Frequency-domain photo thermal radiometry and modulated luminescence (PTR/LUM)	• It is a depth profilometric technique, while other photonic based technologies are not. • Sensitive to changes in both optical and thermal properties of the sample • Has the highest signal dynamic range in detecting very early demineralization. • Reliable and non-invasive	• Signal reception need be improved
Time-correlated single-photon counting fluorescence lifetime imaging (FLIM)	• Relatively safe compared to two photon excitation method.	• Time consuming

6. Endoscopic/Videoscopic Methods

It is established that the difference in the fluorescence of sound and carious enamel is observed after illuminating the tooth with blue light (wavelength 400–500 nm). When this fluoresced tooth is viewed through a specific broadband gelatin filter, white spot lesions appear darker than the surrounding enamel. A white light source can also be connected to an endoscope by a fiberoptic cable so that the teeth can be viewed without a filter. This technique is referred to as 'white light endoscopy.'

Additionally, a camera can be used to store the image. The integration of the camera with the endoscope is called a videoscope. A miniature color video camera is mounted in a custom-made metal mirror holder. This is designed in such a way that the image of the surface of enamel can be viewed directly on a monitor. The video-tapes are viewed by expert examiners who had also examined the teeth by conventional methods.

Advantage

- Facilitate proper viewing (5- to 10-fold magnification achieved).

Disadvantages

- Time consuming (meticulous drying and isolation takes time)
- Costly

7. Dye Penetration Method

Various types of dyes are being used to visualize an object from its routine background or if several objects have a similar appearance, dyes can discriminate them. The observation of the coloring can be qualitative or quantitative. For a qualitative assessment, the colored objects are differentiated from the noncolored ones. For a quantitative assessment, the intensity of color is to be determined. The total area, which is colored, can be compared with the uncolored areas. The intensity of color can be determined by absorption or fluorescence. Absorption measured by quantitating the decrease of light intensity; whereas, fluorescence is measured by quantitating the increase of light intensity at a particular wavelength.

In caries diagnosis, qualitative examination is followed; observation of color change signifies presence of caries. The use of dye is based on the fact that development of microvoids leads to increase in porosity, which is considered as an early sign of caries process. The requisite characteristics of dyes are:

- Should be absolutely safe for intraoral use.
- Should be specific and stain only the tissues it is intended to stain.
- Should be easily removed and not lead to permanent staining.

Dyes used for Detection of Enamel Caries

Various dyes have been tried to detect enamel caries with varying success.

'Procion' dye stains enamel lesions but the staining becomes irreversible because the dye reacts with nitrogen and hydroxyl groups of enamel and acts as a fixative.

'Calcein' dye forms a complex with calcium and remains bound to the lesion. Fluorescent dye (Zyglo ZL-22) has been used *in vitro*, which is not suitable for use *in vivo*. Ultraviolet illumination is required to visualize the dye.

'Brilliant blue' has also been used to enhance the diagnostic quality of fiberoptic transillumination.

Use of dyes for diagnosing enamel lesions cannot be used clinically as yet.

Dyes used for Detection of Dentin Caries

Histopathologically, carious dentin is divided into two layers—outer layer of decalcification, which is soft and cannot be remineralized and the inner decalcified layer, which is hard and can be remineralized. Dyes have been tried to differentiate between these two zones of dentin caries. 0.5% basic fuchsin in propylene glycol has been used for the purpose; however, basic fuchsin dye was proved to be carcinogenic. Later, it was replaced by acid red and methylene blue. Methylene blue is also slightly toxic, so acid red is preferred.

Quantification is not necessary because the dye is only used to identify carious dentin.

Fluorescent dye (Zyglo ZL-22) is used as an aid in diagnosing incipient carious lesions. Zyglo-ZL-22 consists of a solution of chlorinated hydrocarbons and other ingredients, which act as a fluid carrier.

The dye penetration method is modified whereby iodine penetration is used for measuring enamel porosity of the incipient carious lesions. Potassium iodide is applied for a specific period of time to a well-defined area of the enamel and, thereafter, the excess is removed. The iodine, which remains in the micropores, is estimated indicating the permeability of enamel. This method seems complicated because for different areas, a separate calibration is required.

8. Miscellaneous

a. *Laser*

Radiation of wavelength 10.6 μm (commonly used for CO_2 lasers), is strongly absorbed by water and can be vapourized instantaneously. The fact that most

biological tissues contain large quantities of water makes them vulnerable to destruction by irradiation from a laser beam of this wavelength. The water in tissue is vaporized leaving a residue of carbon-based material. The principle of carbon dioxide lasers to be used as a diagnostic tool is based on assumption that subsurface layer of early carious lesion has more organic content when compared with adjacent sound enamel. Photo-vaporization by a CO_2 laser of the organic material in the incipient carious lesion will leave a carbonized residue, which will appear black; the inorganic content of sound enamel with a minimum water content will be less affected.

b. *Infrared Thermography*

The thermal energy emitted by sound tooth structure is compared with that emitted by carious tooth structure. The technique determines lesion activity rather than presence or absence of a lesion. The method uses indium/antimony thermal sensors, which can detect temperature changes in the order of 0.025°C. With a constant flow of air over the surface of the tooth, the change in temperature of the lesion is compared with that of the surrounding sound tooth structure. The source-to-sensor distance is 20 cm, and the time taken to capture the data for a lesion is up to 2 minutes. A reasonable correlation between temperature changes and mineral loss and lesion depth has been established. The technique has not been used *in vivo*, may be because of variations in the temperature of the oral cavity with respiration or fluid evaporation from other oral surfaces. The source-to-specimen distance is presently unsuitable for posterior teeth.

c. *Ultrasonic Imaging*

Ultrasonic imaging the use of sound waves, has been tried to detect early caries in enamel. Ultrasound has the frequency greater than 290 kHz. In contrast to X-ray imaging wherein the image is produced by transmitted radiation; in sonography, the image is produced by the reflected sound waves. The fraction of the beam that is reflected back to the probe is dependent on the acoustic impedance (density and the inherent acoustic velocity) of the tissue to be tested. Ultrasonic beam when directed perpendicular to the surface detects subsurface defects; because the travel time of the sound waves is different in sound and defective surfaces.

Ultrasound waves are produced by an alternating voltage applied to a piezoelectric crystal. Sonic waves that are reflected back may cause a change in the thickness of the piezoelectric crystals, which produces electrical signal. This signal is processed, amplified and stored. Sound waves have to travel through a coupling medium to reach the target, i.e. tooth surface. The ideal coupling agent is one that has acoustic impedance similar to that of the target. A few authors used ultrasound waves, which travel across the surface of the tooth along the interface between enamel and air, rather than through the tooth structure. In this way, ultrasound detects surface discontinuity as a result of cavitated lesions. It is established that ultrasonic caries detector could discriminate interproximal caries from an intact site. Ultrasound may be a quick and reliable tool for the detection of enamel caries. The use of ultrasonic waves to measure demineralization in relation to the cementum-dentin junction is also useful.

Ultrasonics have remained in use for *in vitro* testing studies.

d. *Chemical Analysis*

It is established that nutritional deficiencies are detected by analyzing the chemical composition of teeth. Use of microprobes and other techniques have indicated that trace elements are systematically and inhomogeneously distributed in enamel and dentin. Various trace elements are linked with the carious process. Exposure to environmental cadmium (Cd) has been linked to increased risk of caries. Inductively coupled plasma-mass spectrometry (ICP-MS) has been used to compare the presence/absence or amount of trace elements in healthy vs carious teeth. It is observed that mean concentration of Na, Al, Cr, Mn, Co, Cu, Zn, etc. is in lower quantity in carious teeth as compared to the sound ones; however, Mg, Cd and Pb are observed to be higher in carious teeth than the healthy ones.

e. *Microradiography*

Microradiography is being used to quantify minerals in a given tissue. Three different microradiography techniques are used:

- Transverse microradiography (TMR)
- Longitudinal microradiography (LMR)
- Wavelength independent microradiography (WIM)

f. *Energy Dispersive X-ray Element Analysis*

The mineral content of sound and carious dentin varies. Energy dispersive X-ray element analysis (EDX) is based on the principle that the energy emitted in the form of X-ray photons when the electrons from external sources hit the atoms in a material generating characteristic rays of that element. The change depicts mostly calcium and phosphorus content of dental tissues along with other elements.

g. *Microhardness Analysis*

It is established that early caries lead to dissolution of enamel surface and enlargement of intercrystalline spaces. As the lesion progresses, the surface erosion becomes evident. The affected area is rough, chalky, dull and easily differentiated from the normal enamel. The subsurface demineralization or the surface erosion lead to changes in their hardness characteristics. Cross-sectional microhardness test is used to analyze the tooth with caries (demineralization) as compared to the sound tooth. The microhardness test is performed using Knoop indentron, a microhardness tester (25 gm load is applied for 5 seconds at various angles). Two types of hardness measurement are recognized, (i) where the indentor load is perpendicular to the tooth surface (surface microhardness), and (ii) where indentor load is parallel to the tissues anatomical surface (cross-sectional microhardness). The microhardness test may not differentiate between caries demineralization and erosion due to external factors. Microhardness experiments are mainly carried out on enamel; their use in dentin may cause problems due to relaxation of dentin indentation with time (drying shrinkage effect in dentin).

h. *Iodine Absorptiometry*

Iodine absorptiometry is regarded as a complement to other methods used to analyze mineral density. The advantage of iodine absorptiometry is that the non-destructive measurements can be carried out.

Photon absorptiometry with radioactive iodine and a non-image-forming detector is being used to analyze mineral content of enamel. During mineral loss, the transmission of the radioactive iodine through the mineralized tissue increases, which is recorded with high precision.

i. *Polarized Light Microscopy*

Polarized light microscopy is basically used to analyze mineral loss and gain. The analysis detects the porosity areas of the tooth tissues. (Demineralization leads to more porosity and remineralization means decrease in that porosity.) Polarized light measurements can provide quantitative information on the pore volume (porosity) in the demineralized enamel and also the dentinal lesions.

(For details of caries diagnostic tools, readers are advised to consult the book 'Dental Caries' by Vimal K. Sikri).

BIBLIOGRAPHY

1. Abalos C, Mendoza A, Jmienez-Planas A, Guerrero E, Chaparro A, Garcia-Godoy F. Performance of laser fluorescence for the detection of enamel caries in non-cavitated occlusal surfaces: clinical study with total validation of the sample. Am J Dent 2012;25:44–48.
2. Abdelaziz M, Krejci I, Perneger T, Feilzer A, Vazquez L: Near infrared transillumination compared with radiography to detect and monitor proximal caries: a clinical restrospective study. J Dent 2017;8–16.
3. Alwas Danowska HM, Plasschaert AJ, Suliborski S, Verdonschot EH. Reliability and validity issues of laser fluorescence measurements in occlusal caries diagnosis. J Dent 2002;30:129–34.
4. Ando M, Shaikh S, Eckert G. Determination of caries lesion activity: reflection and roughness for characterization of caries progression. Oper Dent 2018;43:301–06.
5. Ando M, Stookey GK, Zero DT. Ability of quantitative light induced fluorescence (QLF) to assess the activity of white spot lesions during dehydration. Am J Dent 2006;19:15–18.
6. Angmar-Mansson B, Ten Bosch JJ. Advances in methods for diagnosing coronal caries-A review. Adv Dent Res 1993;7:70–79.
7. Angnes V, Angnes G, Batisttella M, Grande RH, Loguercio AD, Reis A. Clinical effectiveness of laser fluorescence, visual inspection and radiography in the detection of occlusal caries. Caries Res 2005;39:490–95.
8. Ansari G, Beeley JA, Reid JS, Foye RH. Caries detector dyes-an *in-vitro* assessment of some new compounds. J. Oral Rehab 1999;26:453–8.
9. Anttonen V, Seppa L, Hausen H. Clinical study of the use of the laser fluorescence device DIAGNOdent for detection of occlusal caries in children. Caries Res 2003; 37:17–23.
10. Arnone D, Ciesla C, Pepper M. Terahertz imaging comes into view. Physics World 2000;13:35–40.
11. Astvaldsdottir A, Holbrook WP, Tranaeus S. Consistency of DIAGNOdent instruments for clinical assessment of fissure caries. Acta Odont Scand 2004;62:193–98.
12. Bader JD, Shugars DA, Bonito AJ. A systematic review of selected dental caries diagnostic and management methods. J Dent Educ 2001;65:960–68.
13. Bader JD, Shugars DA, Bonito AJ. A systematic review of the performance of methods for identifying carious lesions. J Public Health Dent 2002;62:201–13.
14. Bader JD, Shugars DA. A systematic review of the performance of a laser fluorescence device for detecting caries. J Am Dent Assoc 2004;135: 1413–26.
15. Barenie J, Leske G, Ripa LW. The use of fibre optics transillumination for the detection of proximal caries. Oral Surg, Oral Med, Oral Pathol 1973;36:891–97.
16. Bozkurt FO, Tagtekin D, Yanikoglu F, Fontana M, Cabezas CG, Stookey G. Capability of an ultrasonic system to detect very early caries lesions on human enamel. Marmara Dent J 2013; 1:16–19.
17. Brooks SL, Miles DA. Advances in diagnostic imaging in dentistry. Dent Clin North Am 1993;37:91–111.
18. Caliskon Yanikoglu F, Ozturk F, Hayran O, Analoui M, Stookey GK. Detection of natural white spot caries lesions by an ultrasonic system. Caries Res 2000;34:225–32.
19. Chan M, Dadul T, Langlais R, Russell D, Ahmad M. Accuracy of extraoral bite-wing radiography in detecting proximal caries and crestal bone loss. JADA: 2017; 1–7.
20. Ching-Chang Ko, Dong-Ho Yi, Dong Joon L, Jane Kwon, Garcia-Godoy F, Kwon YH. Diagnosis and staging of

caries using spectral factors derived from the blue laser-induced autofluorescence spectrum. J Dent 2017; 1–7.
21. Cortes DF, Ekstrand KR, Elias-Boneta AR, Ellwood RP. An *in-vitro* comparison of the ability of fibre-optic transillumination, visual inspection and radiographs to detect occlusal caries and evaluate lesion depth. Caries Res 2000;34:443–47.
22. Crandell CE, Hill RP. Thermography in dentistry: a pilot study. Oral Surg, Oral Med, Oral Pathol 1966;21:316–20.
23. Crawley DA, Longbottom C, Cole BE, Ciesla CM, Arnone D, Wallace VP, Pepper M. Terahertz pulse imaging: A pilot study of potential application in dentistry. Caries Res 2003;37:352–59.
24. DeJean KS, Caldas LD, Gois DN, Souza CS. Hidden dental caries: a study of diagnosis and prevalence. ClipeOdonto-UNITAU. 2009;1:7–13.
25. Diniz MB, Cordeiro RCL, Ferreira-Zandona AG. Detection of caries around amalgam restorations on approximal surfaces. Oper Dent 2016;41:34–43.
26. Ferreira Zandona A, Zero DT. Diagnostic tools for early caries detection. J Am Dent Assoc 2006;137:1675–84.
27. Fukukita H, Yano T, Fukumoto A. Development and application of an ultrasonic imaging system for dental diagnosis. J Clin Ultrasound 1985;13:597–600.
28. Gimenez T, Piovesan C, Braga MM, Raggio DP, Deery C, Ricketts DN, Ekstrand KR, Mendes FM. Visual Inspection for caries detection: A systematic review and meta-analysis. J Dent Res 2015;94:895–904.
29. Grondahl HG. Digital radiology in dental diagnosis: a critical review. Dentomaxillofacial Radiology: 1992;21:198–202.
30. Gungor K, Erten H, Akarslan ZZ, Celik I, Semiz M. Approximal carious lesions depth assessment with insight and ultraspeed films. Oper Dent 2005;30:58–62.
31. Gupta R, Chaudhary M, Patil S, Bohra S. Energy dispersive X-ray element analysis (EDX) and polarized light microscopic comparison between carious deciduous and permanent teeth. Int J Women Dent 2014;1:43–48.
32. Harase Y, Araki K, Okano T. Accuracy of extraoral tuned operative computed tomography (TACT) for proximal caries detection. Oral Surg., Oral Med, Oral Pathol 2006; 101:791–96.
33. Harorli OT, Barutcigil C, Akgul N, Bayindir YZ. Caries detector dyes: do they stain only the caries? J Rest Dent 2014;2:20–26.
34. Harris R, Nicoll AD, Adair PM, Pine CM. Risk factors for dental caries in young children: a systematic review of the literature. Comm Dent Health 2004;21:71–85.
35. Katkar RA, Tadinada SA, Amaechi BT, Fried D. Optical coherence tomography. Dent Clin N Am 2018; 62:421–34.
36. Kielbassa AM. Current challenges in caries diagnosis. Quint Int 2006;37:421–427.
37. Ko HK, Kang SM, Kim HE, Kwon HK, Kim BI. Validation of Quantitative light-induced fluorescence-digital (QLF-D) for the detection of approximal caries *in vitro*. J Dent 2015;568–572.
38. Kocasarac HD, Angelopoulos C. Ultrasound in dentistry-towards a future of radiation-free imaging. Dent Clin N Am 2018;62:418–89.
39. Kunin AA, Belenova IA, Ippolitov YA, Moiseeva NS, Kunin DA. Predictive research methods of enamel and dentin for initial caries detection. The EPMA Journal 2013;4:19–25.
40. Lees S, Gerhard FB, Oppenheim FG. Ultrasonic measure-ment of dental enamel demineralization. Ultrasonics: 1973;9:269–73.
41. Lennon AM, Buchalla W, Switalski L, Stookey GK. Residual caries detection using visible fluorescence. Caries Res 2002;36:315–19.
42. Lippert F, Lynch RJM. Comparison of Knoop and Vickers surface microhardness and transverse microradiogrpahy for the study of early caries lesion formation in human and bovine enamel. Arch Oral Biol 2014;59:704–10.
43. Longbottom C, Pitts NB, Reich E, Lussi A. Comparison of visual and electrical methods with a new device for occlusal caries detection. Caries Res 1998;32:298.
44. Longbottom C, Pitts NB. CO_2 laser and the diagnosis of occlusal caries: in vitro study. J Dent 1993;21:234–39.
45. Lussi A, Firestone A, Schoenberg V, Hotz P, Stich H. In vivo diagnosis of fissure caries using a new electrical resistance monitor. Caries Res 1995;29:81–87.
46. Lussi A, Francescut P. Performance of conventional and new methods for the detection of occlusal caries in deciduous teeth. Caries Research 2003;37:2–7.
47. Lussi A, Hack A, Hug I, Heckenberger H, Megert B, Stich H. Detection of approximal caries with a new laser fluorescence device. Caries Res 2006;40:97–103.
48. Lussi A, Hibst R, Paulus R. DIAGNOdent: an optical method for caries detection. J Dent Res 83, Spec No. C: C80-C83, 2004.
49. Lussi A, Imwinkelried S, Pitts N, Longbottom C and Reich E. Performance and reproducibility of a laser fluorescence system for detection of occlusal caries in vitro. Caries Res 1999;33:261–66.
50. Lussi A, Megert B, Longbottom C, Reich E, Francescut P. Laser fluorescence may increase diagnostic sensitivity in detecting class I caries. Journal of Evidence-Based Dental Practice 2001;1:95–96.
51. Lussi A. Comparison of different methods for the diagnosis of fissure caries without cavitation. Caries Res 1993;27:409–16.
52. Major IA. Clinical diagnosis of recurrent caries. JADA 2005;136:1426–33.
53. Marsillac MWS, Vieira RS. Assessment of artificial caries lesions through scanning electron microscopy and cross-sectional microhardness test. Indian J Dent Res 2013; 24:249–54.
54. Matos R, Novaes TF, Braga MM, Siqueira WL, Durate DA, Mendes FM. Clinical performance of two fluorescence-based methods in detecting occlusal caries lesions in primary teeth. Caries Res 2011;45:294–302.
55. McComb D, Tam LE. Diagnosis of occlusal caries: Part I. Conventional methods. Journal Canadian Dental Association 2001;67:454–57.
56. Meller C, Heyduck C, Tranaeus S, Splieth C. A new in vitro method for measuring caries activity using quantitative light induced fluorescence. Caries Res 2006;40:90–96.
57. Mitchell JK, Furness AR, Sword RJ, Looney SW, Brackett WW, Brackett MG. Diagnosis of pit and fissure caries using three-dimensional images. Oper Dent 2018; 43:e152–e157.

58. Mohamed A, Helal AF. Analysis of trace elements in teeth by ICP-M: Implications for caries. J Phys Sci 2010;21:1–12.
59. Mount JH. Defining, classifying, and placing incipient caries lesions in perspective. Dent Clin N Am 2005, 49:701–23.
60. Nair MK, Tyndall DA, Ludlow JB, May K. Tuned aperture computed tomography and detection of recurrent caries. Caries Res 1998;32:23–30.
61. Nyvad B. Diagnosis versus detection of caries. Caries Res 2004;38:192–98.
62. Pitts NB, Stamm JW. International consensus workshop on caries clinical trials (ICW-CCT) final consensus statements: agreeing where the evidence leads. J Dent Res 83, Spec No.C: C125-C128, 2004.
63. Pretty IA, Ellwood PG, Davies RM, Worthington HW, Ellwood RP. The effect of illumination and focal distance on light induced fluorescence images *in vitro*. Caries Res 2006;40:73–76.
64. Pretty IA. Caries detection and diagnosis: novel technologies. J Dent 2006;34:727–39.
65. Samarrai SE, Misbah MM. Diagnosis of initial carious lesion by clinical and conventional radiographic methods in comparison to direct digital radiography. Orthodontus, Pedodontus and Preventive Dentistry. J Bajh Coll Dent 2007;19:77–80.
66. Schulze R, Heil U, Gross D. Arefacts in CBCT: a review. Dentomaxillfac Radiol 2011;40:265–73.
67. Stookey GK, Gonzalez-Cabezas C. Emerging methods of caries diagnosis. J Dent Educ 2001; 65:1001–06.
68. Stookey GK. Optical methods-quantitative light fluorescence. J Dent Res 2004;83:C84-C88.
69. Tagliaferro EPS, Junior AV, Rosell FL, Silva SRC, Riley JL, Gilbert GH and Gordon VV. Caries diagnosis in dental practices: Results from dentists in a Brazilian community. Oper Dent 2019;44:E23-E31.
70. Tam LE, Mc Comb D. Diagnosis of occlusal caries: Part II. Recent Diagnostic Technologies. J Can Dent Assoc 2001; 67:459–63.
71. Ten Cate JM. Remineralization of deep enamel dentine caries lesions. Aust Dent J 2008;53:281–85.
72. Tonioli MB, Bouschilcher MR, Hillis SL. Laser fluorescence detection of occlusal caries. Am J Dent 2002;15:268–73.
73. Toraman M, Peker I, Deniz H, Bala O, Altunkaynak B. In vivo comparison of laser fluorescence measurements with conventional methods for occlusal caries detection. Lasers Med Sci 2008;23:307–12.
74. Virajslip V, Thearmontree A, Aryatawong S, Paiboon-warachat D. Comparison of proximal caries detection in primary teeth between laser fluorescence and bitewing radiography. Pediat Dent 2005;27:493–99.
75. Yoon DC, Mol A, Benn DK, Benavides E. Digital radiographic image processing and analysis. Dent Clin N Am 2018; 62:341–59.